

Rewriting Induction for Constrained Inequalities[☆]

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Abstract

Rewriting induction (RI) frameworks consist of inference rules to prove equations to be inductive theorems of a given term rewriting system, i.e., to be inductively valid w.r.t. reduction of the given system. To prove inductive validity of inequalities within such frameworks, one may reduce inequalities to equations. However, it is often hard to prove inductive validity of such reduced equations within the existing RI frameworks due to their indirect handling of inequalities. In this paper, we adapt the notion of inductive theorems to inequalities and propose an RI framework for directly proving inductive validity of inequalities of constrained term rewriting systems. Direct handling of inequalities enables us to utilize transitivity of binary predicates via application of induction hypotheses. Our framework succeeds in proving inductive validity of some inequalities that are hard to verify within the existing RI frameworks for equations. For the sake of automated reasoning, we provide a strategy for applying inference rules and evaluate its performance by means of an implementation.

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1. Introduction

In the field of term rewriting, automated reasoning about *inductive theorems* has been well investigated. Here an inductive theorem of a term rewriting system (TRS) is an equation that is inductively valid, i.e., all of its ground instances are valid w.r.t. reduction of the TRS. As principles for proving inductive theorems, we cite *inductionless induction* [32, 22] and *rewriting induction* (RI) [35], both of which are called *implicit induction principles*. Frameworks based on the RI principle (RI frameworks, for short) consist of inference rules to prove inductive validity, and RI-based methods are procedures within RI frameworks to apply inference rules under specified strategies.

In recent years, various RI-based methods for *constrained rewriting* (see, e.g., constrained TRSs [19, 37], conditional and constrained TRSs [7], $\mathbb{Z}$ -TRSs [16], and logically constrained TRSs [29]) have been developed [7, 37, 16, 30]. Constrained systems have built-in semantics for some function and predicate symbols and have been used as a computational model of not only functional but also imperative programs [14, 18, 19, 15, 40, 30].

While the RI principle has been used for verification of equations, its application to inequality verification has not been widely investigated even though specifications of programs over numbers are sometimes written as inequalities.

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For example, given a unary function f over the integers, we write $f(x) \gtrsim 0$ to indicate that f returns only non-negative integers. The simplest way to prove inductive validity of inequalities within RI frameworks would be to reduce inequalities to equations. Such a reduction, however, does not work well in general. Let us consider verifying inductive validity of a constrained inequality

$$\text{pow}(s^2(0), n) \gtrsim s(n) \quad [n \gtrsim 0] \quad (1)$$

w.r.t. a constrained TRS

$$\mathcal{R}_{\text{pow}} = \left\{ \begin{array}{ll} \text{pow}(x, y) \rightarrow s(0) & [y \not\gtrsim s(0)], \\ \text{pow}(x, y) \rightarrow x \times \text{pow}(x, p(y)) & [y \gtrsim s(0)] \end{array} \right\} \cup \mathcal{R}_{\text{arith}}$$

over a signature containing a binary predicate symbol $\gtrsim$,¹ where $\mathcal{R}_{\text{arith}}$ (shown later) is a set of rewrite rules for arithmetic operators. Here, we suppose that $\gtrsim$ is interpreted as “greater than or equal to” over the integers, and function symbols (e.g., $\times$) except for pow are interpreted by the built-in semantics as the corresponding arithmetic functions, e.g., $s^n(0)$ is interpreted as the integer n . We can reduce inequality (1) to the corresponding equation

$$\text{ge}(\text{pow}(s^2(0), n), s(n)) \approx \text{true} \quad [n \gtrsim 0], \quad (2)$$

where ge is a newly introduced binary symbol representing $\gtrsim$, and we would have to provide additional rewrite rules for ge as follows:

$$\text{ge}(x, y) \rightarrow \text{true} \quad [x \gtrsim y], \quad (3)$$

$$\text{ge}(x, y) \rightarrow \text{false} \quad [x \not\gtrsim y]. \quad (4)$$

In the process of verification based on the RI principle, the initial goal (2) is split into some subgoals, and (2) itself can be used as an induction hypothesis afterward. In our case, we obtain the following equation as one of such subgoals:

$$\text{ge}(s^2(0) \times \text{pow}(s^2(0), p(n')), s(n')) \approx \text{true} \quad [n' \gtrsim 0 \wedge n' \gtrsim s(0)]. \quad (5)$$

The induction hypothesis from (2) ensures that the *inequality* (1) is an induction hypothesis used in, e.g., proving inductive validity of (5). If the subterm $\text{pow}(s^2(0), p(n'))$ were replaced with $s(p(n'))$ by means of (2), verification would be successfully completed because the resulting equation $\text{ge}(s^2(0) \times s(p(n')), s(n')) \approx \text{true} \quad [n' \gtrsim 0 \wedge n' \gtrsim s(0)]$ would have only interpretable symbols except for ge , and thus it can reduce to a valid constraint $n' \geq 1 \Rightarrow 2n' \geq n' + 1$. However, we cannot actually perform such a replacement in, e.g., the framework of [37], because, as an induction hypothesis, (2) actually allows us only to replace an instance of the whole left-hand side of (2) with true . To complete the proof for the example above successfully, we need to allow the partial replacement of $\text{pow}(s^2(0), p(n'))$ with $s(p(n'))$.

While several solutions can be considered to allow the partial replacement including substantial improvement of the existing frameworks (e.g., [37]), one of the most naive but reasonable approaches would be to introduce a framework for *directly* handling inequalities. Direct handling of inequalities enables us to handle the left- and right-hand sides of original inequalities separately, and thus allows the desired partial replacement discussed above. To the best of our knowledge, no implicit-induction-based framework that directly handles inequalities has been developed yet, and therefore it is worth adopting such an approach.

In this paper, we adapt the notion of inductive theorems to inequalities and propose an RI-based framework for directly proving inductive validity of constrained inequalities (Section 3). Basic inference rules for the existing RI frameworks are adapted to handle inequalities (Section 4), e.g., the Expansion rule is split into the L- and R-Expansion rules to distinguish whether the induction hypothesis is applicable from left to right or from right to left. Direct handling of inequalities facilitates the utilization of transitivity of binary predicates. We formalize our RI framework for *many-sorted* systems in order to make our technique theoretically accurate to verification of programs with two or more sorts, e.g., the integers and their lists. To ensure that every ground term is uniquely interpreted under

¹ For readability, we do not abuse $=$ and $\geq$ but use $\approx$ and $\gtrsim$ as predicate symbols in formulas (equalities, inequalities, etc.).

normalization, we require convergence, sufficient completeness, and local soundness for target constrained TRSs. This is because our formalization of inductive validity (i.e., inductive theorems) relies on existence and uniqueness of interpretable normal forms for arbitrary ground terms with interpretable sorts. Although this paper adopts the class of constrained TRSs in [19, 37, 36], it would be straightforward to adapt our results to other higher-level styles of constrained systems in, e.g., [16, 29].

In our approach, inductive validity of inequality (1) in the example above can be proved. Let us glance at how our approach works. Starting with (1), we obtain

$$s^2(0) \times \text{pow}(s^2(0), p(n')) \gtrsim s(n') \quad [n' \gtrsim 0 \wedge n' \gtrsim s(0)] \quad (6)$$

as one of the subgoals, which corresponds to (5). Now we apply (1) as an induction hypothesis to $\text{pow}(s^2(0), p(n'))$ by orienting it from left to right and obtain

$$s^2(0) \times s(p(n')) \gtrsim s(n') \quad [n' \gtrsim 0 \wedge n' \gtrsim s(0)]. \quad (7)$$

Now that (7) has only interpretable function symbols, (7) can be proved to be inductively valid under the built-in semantics. In this way, all the subgoals can be shown to be inductively valid, and thus we succeed in proving the original conjecture (1) is an inductive theorem of $\mathcal{R}_{\text{pow}}$.

Soundness of our approach (shown in Section 5) is based on the fact that if $s \gtrsim t$ is an induction hypothesis and $C[s] \gtrsim u$ is a conjecture to be proved, then verification of $C[t] \gtrsim u$ suffices to prove the conjecture as long as $\gtrsim$ is transitive and $C[\]$ is “positively monotone” (i.e., $v \gtrsim w \Rightarrow C[v] \gtrsim C[w]$ is valid under the built-in semantics for all ground terms v and w). In our example above, the replacement of $\text{pow}(s^2(0), n')$ with $s(n')$ in inequality (6) is sound because $\gtrsim$ is transitive and the context $s^2(0) \times \square$ is “positively monotone” under the built-in semantics. The notion of monotonicity of contexts is not necessary to be considered in the context of equations but needs to be considered in directly handling inequalities. Therefore, we will present a criterion for monotonicity of contexts by showing a sufficient condition.

To automate reasoning within our RI framework, we need a strategy for applying inference rules. Although some strategies have been proposed for equations (see, e.g., [7, 37, 30]), their naive extension does not seem to be suitable without change for our framework handling inequalities instead of equations. One of the reasons is that, in the case of inequalities, application of induction hypotheses may cause approximation of conjectures—it is not always complete—and thus we cannot positively apply induction hypotheses unlike the case of equations. In this paper, based on such observations, we present a strategy designed for our framework, which can prevent excessive application of induction hypotheses, and evaluate its performance by means of an implementation (Section 6). Through experiments, we show that the presented strategy is reasonable for our framework handling inequalities.

Our main contribution is summarized as follows:

- The notion of constrained inequalities is introduced, and the notion of inductive theorems is adapted to constrained inequalities. Constrained inequalities enable us to directly, not by equations, represent inequalities between two uninterpreted terms under the accompanied constraint.
- An RI framework for directly proving inductive validity of constrained inequalities is proposed by making inference rules distinguish the left- and right-hand sides of conjectures and induction hypotheses. The framework proves some conjectures that are hard to verify via reduction of inequalities to equations.
- For the sake of automated reasoning, a strategy for applying inference rules is provided, which is designed for proving constrained inequalities to be inductive theorems. We provide an implementation of our framework coupled with the strategy, and the implementation succeeds in automatically proving some constrained inequalities to be inductive theorems.

Compared to a preliminary work [33], this paper contains the following additional materials:

- We improve the Deletion rule to strictly enhance the power of the framework.
- We develop a strategy for applying inference rules and evaluate their performance through experiments.

- All the proofs for soundness of our inference rules are included.
- We formalize the framework in the context of many-sorted systems because it is difficult to achieve sufficient completeness in single-sorted systems which handle not only the integers but also other data structures.
- To prove ground confluence, we slightly modify the examples by removing interpretable function symbols from the left-hand sides of rewrite rules for uninterpreted function symbols. This change does not affect discussions in the paper.

This paper is organized as follows. Section 2 recalls the basic notions and notations used in the paper. Section 3 introduces the notion of constrained inequalities and adapts the notion of inductive theorems to inequalities. In Section 4, we present our framework for proving inequalities to be inductive theorems. A criterion for monotonicity of contexts is also presented in this section. Section 5 proves soundness of our inference rules. Section 6 discusses automation of our framework and shows experimental results via an implementation of our framework. Section 7 describes related work. In Section 8, we conclude this paper and discuss future work.

2. Preliminaries

In this section, we briefly recall some basic notions and notations of term rewriting [5], the first-order predicate logic [23], and constrained term rewriting systems [37]. To employ first-order many-sorted terms in this paper, we extend some notions into many-sorted ones in a natural way [26].

2.1. Terms, contexts, and substitutions

Let $\mathcal{S}$ be a set of *basic sorts* and $\mathcal{V} (= \bigcup_{\tau \in \mathcal{S}} \mathcal{V}_\tau)$ be a set of many-sorted *variables* over $\mathcal{S}$, where the sets $\mathcal{V}_\tau$ are countably infinite and pairwise disjoint. Let Σ be a many-sorted *signature*, namely a set of many-sorted *function symbols* over $\mathcal{S}$. Each function symbol f is equipped with a sort $\tau_1 \times \cdots \times \tau_n \rightarrow \tau$ and is denoted by $f : \tau_1 \times \cdots \times \tau_n \rightarrow \tau$, where $\tau_1, \dots, \tau_n, \tau \in \mathcal{S}$ and $n \geq 0$. The set of *well-sorted terms* over Σ and $\mathcal{V}$ is denoted by $\mathcal{T}(\Sigma, \mathcal{V})$, and the set of *ground terms* over Σ is denoted by $\mathcal{T}(\Sigma)$, where we consider only well-sorted terms as usual. The notation $t : \tau$ denotes a term t of sort τ . The set of variables occurring in the terms $t_1, \dots, t_n$ is denoted by $\text{Var}(t_1, \dots, t_n)$. A term t is said to be *linear* if no variable appears more than once in t . For a unary function symbol $f : \tau \rightarrow \tau$, the notation $f^n(t)$ abbreviates the n -fold application $f(\cdots f(t) \cdots)$ of f to t . The *identity* over terms is denoted by $\equiv$. For a term t and a *position* p of t , the notation $t|_p$ represents the *subterm* of t at p . For a non-variable term t , the notation $\text{root}(t)$ represents the *root symbol* of t .

For a *context* $C[\]_p$ and a term t , the notation $C[t]_p$ represents the term obtained by replacing the *hole* $\square_\tau$ of sort τ at the position p in $C[\]$ with t of sort τ , where $C[t]_p$ must be well-sorted. The hole itself is a context, the *trivial* context. The set of contexts over Σ and $\mathcal{V}$ is denoted by $\mathcal{T}_\square(\Sigma, \mathcal{V})$.

A *substitution* is a sort-preserving mapping from $\mathcal{V}$ to $\mathcal{T}(\Sigma, \mathcal{V})$. The application $t\sigma$ of a substitution σ to a term t yields the term obtained by replacing each variable x in t with $\sigma(x)$. A term s is called an *instance* of t (denoted by $s \succsim t$) if $s \equiv t\sigma$ for some substitution σ . The *domain* $\text{Dom}(\sigma)$ of σ is the set of variables x such that $x \neq \sigma(x)$, and the *range* $\text{Ran}(\sigma)$ of σ is $\{\sigma(x) \mid x \in \text{Dom}(\sigma)\}$. A substitution σ whose domain is $\{x_1, \dots, x_n\}$ can be written as $\{x_1 \mapsto \sigma(x_1), \dots, x_n \mapsto \sigma(x_n)\}$. The *restriction* $\sigma|_X$ of σ to $X \subseteq \mathcal{V}$ is $\sigma|_X = \{x \mapsto \sigma(x) \mid x \in \text{Dom}(\sigma) \cap X\}$. The composition $\sigma\theta$ of substitutions σ and θ is defined by $(\sigma\theta)(x) = \theta(\sigma(x))$. A substitution σ is said to *ground* terms $t_1, \dots, t_n$ (and is called a *ground substitution* w.r.t. $t_1, \dots, t_n$) if $\text{Dom}(\sigma) \supseteq \text{Var}(t_1, \dots, t_n)$ and $\text{Ran}(\sigma|_{\text{Var}(t_1, \dots, t_n)}) \subseteq \mathcal{T}(\Sigma)$. A *most general unifier* (mgu) of terms s and t is denoted by $\text{mgu}(s, t)$ if it exists. Note that an mgu of two terms is unique up to renaming.

2.2. First-order predicate logic

Let $\mathcal{S}$ be a set of sorts, $\mathcal{G}$ be a many-sorted signature over $\mathcal{S}$ (e.g., a subsignature of Σ), and $\mathcal{P}$ be a finite set of many-sorted *predicate symbols* over $\mathcal{S}$. Each predicate symbol p is equipped with a sort $\tau_1 \times \cdots \times \tau_n$ and is denoted by $p : \tau_1 \times \cdots \times \tau_n$, where $\tau_1, \dots, \tau_n \in \mathcal{S}$ and $n \geq 0$. *Formulas* over $(\mathcal{G}, \mathcal{P}, \mathcal{V})$ are defined in BNF as follows: $\phi ::= p(t_1, \dots, t_n) \mid \top \mid \perp \mid (\neg\phi) \mid (\phi \wedge \phi) \mid (\phi \vee \phi) \mid (\phi \Rightarrow \phi) \mid (\forall x. \phi) \mid (\exists x. \phi)$, where $p : \tau_1 \times \cdots \times \tau_n \in \mathcal{P}$ and $t_1 : \tau_1, \dots, t_n : \tau_n \in \mathcal{T}(\mathcal{G}, \mathcal{V})$. We omit parentheses in formulas in the usual way. The set of *free variables* in a

formula ϕ is denoted by $FV(\phi)$. A formula is said to be *closed* if it has no free variables. A substitution σ from $\mathcal{V}$ to $\mathcal{T}(\Sigma, \mathcal{V})$ with $\text{Ran}(\sigma|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G}, \mathcal{V})$ can be applied to a formula ϕ in the usual way. A substitution σ is said to *ground* formulas $\phi_1, \dots, \phi_n$ if $\text{Dom}(\sigma) \supseteq \bigcup_{1 \leq i \leq n} FV(\phi_i)$ and $\text{Ran}(\sigma|_{\bigcup_{1 \leq i \leq n} FV(\phi_i)}) \subseteq \mathcal{T}(\Sigma)$.

A *structure* $\mathcal{M}$ for $(\mathcal{S}, \mathcal{G}, \mathcal{P})$ is a triple $(\mathcal{A}, \mathcal{I}_{\mathcal{G}}, \mathcal{I}_{\mathcal{P}})$, where $\mathcal{A} = \{\mathcal{A}_{\tau} \mid \tau \in \mathcal{S}\}$ is a set of *universes* (namely non-empty sets of concrete values), $\mathcal{I}_{\mathcal{G}}$ is a mapping from $\mathcal{G}$ to functions such that $\mathcal{I}_{\mathcal{G}}(g) : \mathcal{A}_{\tau_1} \times \dots \times \mathcal{A}_{\tau_n} \rightarrow \mathcal{A}_{\tau}$ for every $g : \tau_1 \times \dots \times \tau_n \rightarrow \tau \in \mathcal{G}$, and $\mathcal{I}_{\mathcal{P}}$ is a mapping from $\mathcal{P}$ to sets of value tuples such that $\mathcal{I}_{\mathcal{P}}(p) \subseteq \mathcal{A}_{\tau_1} \times \dots \times \mathcal{A}_{\tau_n}$ for every $p : \tau_1 \times \dots \times \tau_n \in \mathcal{P}$. The *interpretation* $(g(t_1, \dots, t_n))^{\mathcal{M}}$ of a ground term $g(t_1, \dots, t_n)$ is inductively defined by $\mathcal{I}_{\mathcal{G}}(g)(t_1^{\mathcal{M}}, \dots, t_n^{\mathcal{M}})$. A closed atomic formula $p(t_1, \dots, t_n)$ is said to *hold* in $\mathcal{M}$, written as $\mathcal{M} \models p(t_1, \dots, t_n)$, if $(t_1^{\mathcal{M}}, \dots, t_n^{\mathcal{M}}) \in \mathcal{I}_{\mathcal{P}}(p)$. The relation $\models$ is naturally extended to arbitrary closed formulas in the usual way. A formula ϕ is said to be *$\mathcal{M}$ -valid* (resp. *$\mathcal{M}$ -satisfiable*) if $\mathcal{M} \models \phi\sigma$ for every (resp. some) substitution σ grounding ϕ . When $\mathcal{G}$, $\mathcal{P}$, and $\mathcal{M}$ are fixed in the context, we may call formulas over $(\mathcal{G}, \mathcal{P}, \mathcal{V})$ *constraints* (w.r.t. $\mathcal{M}$). We suppose that for every sort $\tau \in \mathcal{S}$ and element $a \in \mathcal{A}_{\tau}$, there exists a ground term $t : \tau$ such that $t^{\mathcal{M}} = a$. This implies that, for each sort $\tau \in \mathcal{S}$, $\mathcal{G}$ contains a constant a of the sort τ . We also suppose that $\mathcal{P}$ contains a binary predicate symbol $\simeq_{\tau}$ for each sort $\tau \in \mathcal{S}$ whose interpretation $\simeq_{\tau}^{\mathcal{M}}$ is the equality predicate over $\mathcal{A}_{\tau}$.

2.3. Constrained TRSs

Let $\mathcal{S}_{\mathcal{U}}$ and $\mathcal{S}_{\mathcal{I}}$ be disjoint sets of basic sorts (i.e., $\mathcal{S} = \mathcal{S}_{\mathcal{U}} \uplus \mathcal{S}_{\mathcal{I}}$)— $\mathcal{S}_{\mathcal{I}}$ is the set of sorts for interpreted symbols, and $\mathcal{S}_{\mathcal{U}}$ is for others— $\mathcal{F}$ be a signature over $\mathcal{S}_{\mathcal{U}} \cup \mathcal{S}_{\mathcal{I}}$, $\mathcal{G}$ be a signature over $\mathcal{S}_{\mathcal{I}}$ disjoint from $\mathcal{F}$ (i.e., $\Sigma = \mathcal{F} \uplus \mathcal{G}$),² $\mathcal{P}$ be a set of predicate symbols over $\mathcal{S}_{\mathcal{I}}$, $\mathcal{M}$ be a structure for $(\mathcal{S}_{\mathcal{I}}, \mathcal{G}, \mathcal{P})$, where we assume that there exists at least one constant symbol for each basic sort. A *constrained rewrite rule* over $(\mathcal{S}_{\mathcal{U}}, \mathcal{S}_{\mathcal{I}}, \mathcal{F}, \mathcal{G}, \mathcal{P}, \mathcal{M})$ is a triple (l, r, ϕ) , written as $l \rightarrow r \ [\phi]$, such that l and r are terms in $\mathcal{T}(\mathcal{F} \cup \mathcal{G}, \mathcal{V})$ of the same sort, l is not a variable, ϕ is a constraint over $(\mathcal{G}, \mathcal{P}, \mathcal{V})$, $\text{Var}(r) \cup FV(\phi) \subseteq \text{Var}(l)$, and ϕ is $\mathcal{M}$ -satisfiable. We may write $l \rightarrow r$ instead of $l \rightarrow r \ [\top]$.

Let $\mathcal{R}$ be a finite set of constrained rewrite rules over $(\mathcal{S}_{\mathcal{U}}, \mathcal{S}_{\mathcal{I}}, \mathcal{F}, \mathcal{G}, \mathcal{P}, \mathcal{M})$. The *rewrite relation* $\rightarrow_{\mathcal{R}}$ of $\mathcal{R}$ is defined by $\rightarrow_{\mathcal{R}} = \{(C[l\sigma]_p, C[r\sigma]_p) \mid l \rightarrow r \ [\phi] \in \mathcal{R}, C[\] \in \mathcal{T}_{\square}(\mathcal{F} \cup \mathcal{G}, \mathcal{V}), \text{Ran}(\sigma|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G}, \mathcal{V}), \phi\sigma \text{ is } \mathcal{M}\text{-valid}\}$, where $l\sigma$ in the definition of $\rightarrow_{\mathcal{R}}$ is called a *redex* of $\mathcal{R}$. A *constrained term rewriting system* (constrained TRS) over $(\mathcal{S}_{\mathcal{U}}, \mathcal{S}_{\mathcal{I}}, \mathcal{F}, \mathcal{G}, \mathcal{P}, \mathcal{M})$ is an *abstract reduction system* $(\mathcal{T}(\mathcal{F} \cup \mathcal{G}, \mathcal{V}), \rightarrow_{\mathcal{R}})$, simply written as $\mathcal{R}$. Note that $\mathcal{R}$ is a *term rewriting system* (TRS) if $\mathcal{S}_{\mathcal{U}} \cup \mathcal{S}_{\mathcal{I}}$ is a singleton and all the constraints of the rewrite rules in $\mathcal{R}$ are $\top$. The restriction $\mathcal{R}|_F$ of $\mathcal{R}$ to a subsignature $F \subseteq \mathcal{F} \cup \mathcal{G}$ is $\{l \rightarrow r \ [\phi] \in \mathcal{R} \mid \text{root}(l) \in F\}$. A term t is said to be *reducible* if there exists a term t' such that $t \rightarrow_{\mathcal{R}} t'$. Otherwise, t is *irreducible* (or *in normal form*). A term t is called a *normal form* of s (w.r.t. $\mathcal{R}$) if $s \rightarrow_{\mathcal{R}}^* t$ and t is irreducible. A substitution is said to be *normalized* if its range consists of only normal forms. A position p of a term t is said to be *reduction-complete* under a constraint ϕ if, for any normalized substitution σ grounding $t|_p$ with $\text{Ran}(\sigma|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G}, \mathcal{V})$, the subterm $(t\sigma)|_p$ of $t\sigma$ at p is a redex of $\mathcal{R}$, whenever $\phi\sigma$ is $\mathcal{M}$ -valid.³

Example 2.1. Let *int* and *list* be sorts for interpreted and uninterpreted symbols, respectively. Let $\mathcal{G}_{\mathbb{Z}} = \{0 : \text{int}, s : \text{int} \rightarrow \text{int}, p : \text{int} \rightarrow \text{int}, + : \text{int} \times \text{int} \rightarrow \text{int}, - : \text{int} \times \text{int} \rightarrow \text{int}, \times : \text{int} \times \text{int} \rightarrow \text{int}\}$, $\mathcal{P}_{\mathbb{Z}} = \{\approx : \text{int} \times \text{int}, \gtrsim : \text{int} \times \text{int}\}$, and $\mathcal{M}_{\mathbb{Z}}$ be a structure for $(\{\text{int}\}, \mathcal{G}_{\mathbb{Z}}, \mathcal{P}_{\mathbb{Z}})$ where the universe of *int* consists of the integers $\mathbb{Z}$ and function symbols are interpreted as follows: $0^{\mathcal{M}_{\mathbb{Z}}} = 0$, $s^{\mathcal{M}_{\mathbb{Z}}}(x) = x + 1$, $p^{\mathcal{M}_{\mathbb{Z}}}(x) = x - 1$, $+^{\mathcal{M}_{\mathbb{Z}}}(x, y) = x + y$, $-^{\mathcal{M}_{\mathbb{Z}}}(x, y) = x - y$, and $\times^{\mathcal{M}_{\mathbb{Z}}}(x, y) = x \times y$. We assume that the predicate symbols $\approx$ and $\gtrsim$ in $\mathcal{P}_{\mathbb{Z}}$ are interpreted as the corresponding operators $=$ and $\geq$ over the integers, respectively, and we use $\neq$ and $\not\geq$ as syntactic sugars for the negations of $\approx$ and $\gtrsim$, respectively. Let us consider the following constrained TRS $\mathcal{R}_{\text{count}}$ over $(\{\text{list}\}, \{\text{int}\}, \{\text{nil} : \text{list}, \text{cons} : \text{int} \times \text{list} \rightarrow \text{list}, \text{count} : \text{int} \times \text{list} \rightarrow \text{list}, \text{length} : \text{list} \rightarrow \text{int}\}, \mathcal{G}_{\mathbb{Z}}, \mathcal{P}_{\mathbb{Z}}, \mathcal{M}_{\mathbb{Z}})$:

$$\mathcal{R}_{\text{count}} = \left\{ \begin{array}{l} \text{count}(x, \text{nil}) \rightarrow 0, \\ \text{count}(x, \text{cons}(y, ys)) \rightarrow s(\text{count}(x, ys)) \quad [x \approx y], \\ \text{count}(x, \text{cons}(y, ys)) \rightarrow \text{count}(x, ys) \quad [x \neq y], \\ \text{length}(\text{nil}) \rightarrow 0, \\ \text{length}(\text{cons}(x, xs)) \rightarrow s(\text{length}(xs)) \end{array} \right\} \cup \mathcal{R}_{\text{arith}},$$

²A signature Σ is explicitly divided into $\mathcal{F}$ and $\mathcal{G}$, where $\mathcal{F}$ is the set of *uninterpreted* symbols and $\mathcal{G}$ is the set of *interpreted* symbols.

³Reduction completeness is introduced in [17] and is originally called *completeness*.

where $\mathcal{R}_{\text{arith}}$ is a set of rewrite rules for arithmetic operators defined by

$$\mathcal{R}_{\text{arith}} = \left\{ \begin{array}{lll} x + 0 \rightarrow x, & 0 \times y \rightarrow 0, & x - 0 \rightarrow x, \\ x + \mathbf{s}(y) \rightarrow \mathbf{s}(x + y), & \mathbf{s}(x) \times y \rightarrow (x \times y) + y, & x - \mathbf{s}(y) \rightarrow \mathbf{p}(x - y), \\ x + \mathbf{p}(y) \rightarrow \mathbf{p}(x + y), & \mathbf{p}(x) \times y \rightarrow (x \times y) - y, & x - \mathbf{p}(y) \rightarrow \mathbf{s}(x - y), \\ \mathbf{s}(\mathbf{p}(x)) \rightarrow x, & \mathbf{p}(\mathbf{s}(x)) \rightarrow x & \end{array} \right\}.$$

This constrained TRS $\mathcal{R}_{\text{count}}$ defines functions counting elements of lists: `count` counts the number of elements in a given list as its second argument, which are equivalent to a given element as its first argument; `length` counts the number of elements in a given list as its first argument. For example, letting $t \equiv \text{cons}(0, \text{cons}(0, \text{cons}(\mathbf{s}(0), \text{nil})))$, we have that $\text{count}(t, 0) \rightarrow_{\mathcal{R}_{\text{count}}}^* \mathbf{s}^2(0)$ and $\text{length}(t) \rightarrow_{\mathcal{R}_{\text{count}}}^* \mathbf{s}^3(0)$.

A constrained TRS $\mathcal{R}$ is said to be (*ground*) *terminating* if there exists no infinite sequence $t_1 \rightarrow_{\mathcal{R}} t_2 \rightarrow_{\mathcal{R}} \dots$ of (ground) terms,⁴ and $\mathcal{R}$ is (*ground*) *confluent* if $t_1 \leftarrow_{\mathcal{R}}^* t \rightarrow_{\mathcal{R}}^* t_2$ implies $t_1 \rightarrow_{\mathcal{R}}^* \circ \leftarrow_{\mathcal{R}}^* t_2$ for all (ground) terms t, t_1 , and t_2 . We say that $\mathcal{R}$ is *ground convergent* if it is both ground terminating and ground confluent. If $\mathcal{R}$ is ground convergent, the normal form of a ground term t is unique and denoted by $t \downarrow_{\mathcal{R}}$. A constrained TRS $\mathcal{R}$ is said to be *sufficiently complete w.r.t. interpretable terms* if, for any ground term $s : \tau \in \mathcal{T}(\mathcal{F} \cup \mathcal{G})$ with $\tau \in \mathcal{S}_I$, there exists a term $t : \tau \in \mathcal{T}(\mathcal{G})$ such that $s \rightarrow_{\mathcal{R}}^* t$.⁵ A constrained TRS $\mathcal{R}$ is said to be *quasi-reductive w.r.t. interpretable terms* if every ground term in $\mathcal{T}(\mathcal{F} \cup \mathcal{G}) \setminus \mathcal{T}(\mathcal{G})$ of sort $\tau \in \mathcal{S}_I$ is reducible. By definition, it is clear that every terminating constrained TRS which is quasi-reductive w.r.t. interpretable terms is sufficient complete w.r.t. interpretable terms, i.e., sufficient completeness and quasi-reductivity are equivalent in terminating constrained TRSs. A constrained TRS $\mathcal{R}$ is said to be *locally sound w.r.t. $\mathcal{M}$* if, for every ground term $s : \tau$ in $\mathcal{T}(\mathcal{G})$, $s \rightarrow_{\mathcal{R}} t$ implies that $t \in \mathcal{T}(\mathcal{G})$ and $\mathcal{M} \models s \simeq_{\tau} t$. A constrained TRS $\mathcal{R}$ is said to be *left-linear* if the left-hand side of every rewrite rule in $\mathcal{R}$ is linear.

Throughout the paper, $\mathcal{S}_U$ and $\mathcal{S}_I$ denote disjoint sets of basic sorts, $\mathcal{V}$ an infinite set of variables, $\mathcal{F}$ a signature over $\mathcal{S}_U \cup \mathcal{S}_I$, $\mathcal{G}$ a signature over $\mathcal{S}_I$ disjoint from $\mathcal{F}$, $\mathcal{P}$ a set of predicate symbols over $\mathcal{S}_I$, $\mathcal{M}$ a structure for $(\mathcal{S}_I, \mathcal{G}, \mathcal{P})$, and $\mathcal{R}$ a constrained TRS over $(\mathcal{S}_U, \mathcal{S}_I, \mathcal{F}, \mathcal{G}, \mathcal{P}, \mathcal{M})$. To ensure that every ground term is uniquely interpreted, we assume that $\mathcal{R}$ is ground convergent, sufficiently complete w.r.t. interpretable terms, and locally sound w.r.t. the associated structure $\mathcal{M}$ unless noted otherwise.

2.4. Existing methods to prove ground convergence, local soundness, and sufficient completeness

There exists a series of works to prove termination, (ground) confluence, sufficient completeness, and local soundness of constrained TRSs. Regarding termination, dependency pair frameworks adapted for rewriting systems with built-in semantics are developed in [14, 18, 15, 36, 28]. Termination of all the examples in this paper can be verified by the method of [36], although we will omit the details. To confirm (ground) confluence of terminating constrained TRSs, it suffices to show that all their *feasible* constrained critical pairs are joinable [19]. Local soundness w.r.t. the associated structure can be decided by a necessary and sufficient condition presented in [37, Lemmas 3.5 and 3.6]: if for every rewrite rule $l \rightarrow r$ [ϕ] $\in \mathcal{R}$, $l \in \mathcal{T}(\mathcal{G}, \mathcal{V})$ implies $r \in \mathcal{T}(\mathcal{G}, \mathcal{V})$ and $\mathcal{M}$ -validity of $\phi \Rightarrow l \simeq r$, then $\mathcal{R}$ is locally sound w.r.t. $\mathcal{M}$. For terminating TRSs, sufficient completeness w.r.t. interpretable terms can be reduced to quasi-reductivity w.r.t. interpretable terms, and for left-linear terminating constrained TRSs, the sufficient completeness can be verified by a natural extension of test set techniques [24, 25]. Assume that a subsignature $\mathcal{C} \subseteq \mathcal{F} \cup \mathcal{G}$ is given, where $c : \tau \in \mathcal{C}$ with $\tau \in \mathcal{S}_I$ implies $c \in \mathcal{G}$. Then, quasi-reductivity w.r.t. interpretable terms is ensured by “quasi-reductivity w.r.t. $\mathcal{C}$ ”, where we say that $\mathcal{R}$ is *quasi-reductive w.r.t. $\mathcal{C}$* if every term in $\mathcal{T}(\mathcal{F} \cup \mathcal{G}) \setminus \mathcal{T}(\mathcal{C})$ is reducible. A constrained TRS $\mathcal{R}$ is proved to be quasi-reductive w.r.t. $\mathcal{C}$ if $\bigvee_{\phi \in \text{Const}(t)} \phi$ is $\mathcal{M}$ -valid for every function symbol $f \in (\mathcal{F} \cup \mathcal{G}) \setminus \mathcal{C}$ and every term $t \equiv f(t_1, \dots, t_n)$, where $\text{Const}(t) = \{\phi\sigma \mid l \rightarrow r [\phi] \in \mathcal{R}, l \preceq t, l\sigma \equiv t\}$, and $t_1, \dots, t_n$ are renamed variants of terms in a test set of the TRS $\{l \rightarrow r \mid l \rightarrow r [\phi] \in \mathcal{R}\}$ w.r.t. $\mathcal{C}$. Although there may be multiple candidates for the subsignature $\mathcal{C}$, it suffices to try finitely many cases to find appropriate one, and typically we let $\mathcal{C} = (\mathcal{F} \cup \mathcal{G}) \setminus (\text{Root}(\mathcal{R}) \setminus \text{Inner}(\mathcal{R}))$, where $\text{Root}(\mathcal{R})$ denotes $\{\text{root}(l) \mid l \rightarrow r [\phi] \in \mathcal{R}\}$ and $\text{Inner}(\mathcal{R})$ denotes the set of symbols appearing in the direct subterms of left-hand sides of $\mathcal{R}$.

⁴Note that termination and ground termination are equivalent in our setting because each sort has at least one constant.

⁵This is a straightforward extension of sufficient completeness w.r.t. constructor terms [21] to interpretable terms.

Example 2.2. Using these criteria, we can confirm that the constrained TRS $\mathcal{R}_{\text{isqrt}}$ in Example 3.3 is ground convergent, sufficiently complete w.r.t. interpretable terms, and locally sound w.r.t. $\mathcal{M}_{\mathbb{Z}}$. Termination of $\mathcal{R}_{\text{isqrt}}$ is confirmed by the methods mentioned in this subsection. The constrained TRS $\mathcal{R}_{\text{isqrt}}$ is ground confluent because critical pairs occur only in $\mathcal{R}_{\text{arith}}$ and it is easy to see that all of them are joinable on ground terms. The constrained TRS $\mathcal{R}_{\text{isqrt}}$ is locally sound w.r.t. $\mathcal{M}_{\mathbb{Z}}$ because it satisfies the condition mentioned in this subsection. Letting $C = \{0, s, p\}$, we obtain a test set $\{0, s(0), s(s(x)), p(0), p(p(x))\}$ by the method in [25] with ignoring constraints of $\mathcal{R}_{\text{isqrt}}$. With this test set, sufficient completeness w.r.t. interpretable terms can easily be proved along the way shown in this subsection. For example, letting t be $u(0, s(s(x)))$, we have that $\bigvee_{\phi \in \text{Const}(t)} \phi \equiv s(s(x)) \gtrsim s(0) \times s(0) \vee s(s(x)) \not\gtrsim s(0) \times s(0)$, which is $\mathcal{M}_{\mathbb{Z}}$ -valid.

3. Constrained inequalities

In this section, we formalize the notion of constrained inequalities. This allows us to directly describe an inequality between two uninterpreted terms under the attached constraint. In Section 3.1, in addition to formalization of constrained inequalities, adaptation of inductive theorems to constrained inequalities is presented. In Section 3.2, we present a sufficient condition of trivial inductive theorems. In the remainder of the paper, we assume that $\mathcal{R}$ has a binary predicate symbol of sort $\tau \times \tau$ in $\mathcal{P}$ that forms inequalities (e.g., $\gtrsim$ in Example 2.1).

3.1. Formalization of constrained inequalities and inductive theorems

First, we define a constrained inequality. This represents a relation, typically some magnitude relation, between two terms under the attached constraint. For instance, one of the specifications of a function `isqrt` (shown later) calculating the integer square root can be represented by a constrained inequality $n \gtrsim \text{isqrt}(n) \times \text{isqrt}(n)$ [$n \gtrsim 0$], which means that $n \gtrsim 0$ implies $n \gtrsim \text{isqrt}(n) \times \text{isqrt}(n)$ for all integers n . Constrained inequalities are formalized as follows:

Definition 3.1. Let $\gtrsim : \tau \times \tau$ be a binary predicate symbol in $\mathcal{P}$. A *constrained $\gtrsim$ -inequality* is a quadruple $(\gtrsim, s, t, \phi)$ of $\gtrsim$, terms s and t in $\mathcal{T}(\mathcal{F} \cup \mathcal{G}, \mathcal{V})$ of sort τ , and a constraint ϕ over $(\mathcal{G}, \mathcal{P}, \mathcal{V})$. For readability, we write $s \gtrsim t$ [ϕ] for $(\gtrsim, s, t, \phi)$.

When $\gtrsim$ is clear from the context, we simply write “constrained inequalities” instead of “constrained $\gtrsim$ -inequalities”. We do not impose any variable condition such as $FV(\phi) \subseteq \text{Var}(s, t)$ unlike rewrite rules. When we interpret $\gtrsim$ by $>$, a $\gtrsim$ -inequality $s \gtrsim t$ represents an inequality of the form $s > t$.

Next, we define inductive validity of a constrained inequality w.r.t. reduction of a given constrained TRS. As in the case of equations, inductively valid constrained inequalities are called inductive theorems of the constrained TRS.

Definition 3.2. A constrained inequality $s \gtrsim t$ [ϕ] is called an *inductive theorem* of $\mathcal{R}$ if $\mathcal{M} \models s\sigma \downarrow_{\mathcal{R}} \gtrsim t\sigma \downarrow_{\mathcal{R}}$ for every substitution σ grounding s and t such that $\text{Ran}(\sigma|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G}, \mathcal{V})$ and $\phi\sigma$ is $\mathcal{M}$ -valid.

Example 3.3. Consider int , $G_{\mathbb{Z}}$, $\mathcal{P}_{\mathbb{Z}}$, $\mathcal{M}_{\mathbb{Z}}$, and $\mathcal{R}_{\text{arith}}$ in Example 2.1 again. Let us consider the following constrained TRS $\mathcal{R}_{\text{isqrt}}$ over $(\emptyset, \{\text{int}\}, \{\text{isqrt} : \text{int} \rightarrow \text{int}, u : \text{int} \times \text{int} \rightarrow \text{int}\}, G_{\mathbb{Z}}, \mathcal{P}_{\mathbb{Z}}, \mathcal{M}_{\mathbb{Z}})$:

$$\mathcal{R}_{\text{isqrt}} = \left\{ \begin{array}{ll} \text{isqrt}(n) \rightarrow u(0, n), & \\ u(i, n) \rightarrow u(s(i), n) & [n \gtrsim s(i) \times s(i)], \\ u(i, n) \rightarrow i & [n \not\gtrsim s(i) \times s(i)] \end{array} \right\} \cup \mathcal{R}_{\text{arith}}.$$

The constrained TRS $\mathcal{R}_{\text{isqrt}}$ defines a function `isqrt` calculating the integer square root. For example, $\text{isqrt}(s^{10}(0))$ evaluates to $s^3(0)$. The following constrained inequalities represent the specification of the integer square root function:

$$\begin{array}{ll} \text{isqrt}(n) \gtrsim 0 & [n \gtrsim 0], \\ n \gtrsim \text{isqrt}(n) \times \text{isqrt}(n) & [n \gtrsim 0], \\ s(\text{isqrt}(n)) \times s(\text{isqrt}(n)) \gtrsim s(n) & [n \gtrsim 0]. \end{array}$$

In Section 4.4, using an RI framework for inequalities, we will show that these inequalities are inductive theorems of $\mathcal{R}_{\text{isqrt}}$.

3.2. Trivial inductive theorems

Within RI frameworks, removal of trivial inductive theorems is a key operation to decrease the number of subgoals. In this subsection, we revisit sufficient conditions of trivial inductive theorems.

A typical sufficient condition of trivial inductive theorems of the form $s \approx t [\phi]$ in former frameworks is that $s \equiv t$ or ϕ is unsatisfiable. Since inductive validity of inequalities is defined by the built-in semantics of a given constrained TRS, the condition is modified for a constrained inequality $s \gtrsim t [\phi]$ as follows: the constraint $\phi \Rightarrow s' \gtrsim t'$ is valid w.r.t. the built-in semantics, where s' (resp. t') is a term obtained by replacing uninterpretable subterms of s (resp. t) with fresh variables. In this replacement, it is expected that the same uninterpretable subterms are replaced with the same variable. For example, the terms $\mathbf{s}(\mathbf{u}(i, n))$ and $\mathbf{u}(i, n)$ are expected to be transformed into $\mathbf{s}(x)$ and x , respectively.

For this purpose, we define the “floor” $\lfloor (t_1, \dots, t_n) \rfloor_{\mathcal{G}}$ of terms $(t_1, \dots, t_n)$. This represents terms obtained by replacing each uninterpretable subterm of $t_1, \dots, t_n$ by a fresh variable. In the context of the dependency pair framework [4], a similar function is known by the name “CAP”, which replaces non-constructor subterms of a given term with variables. The operation $\lfloor \cdot \rfloor_{\mathcal{G}}$ is defined as follows.

Definition 3.4. Given an n -tuple $(t_1, \dots, t_n)$ of terms in $\mathcal{T}(\mathcal{F} \cup \mathcal{G}, \mathcal{V})$, we define the n -tuple $\lfloor (t_1, \dots, t_n) \rfloor_{\mathcal{G}}$ of terms in $\mathcal{T}(\mathcal{G}, \mathcal{V})$ by

$$\lfloor (t_1, \dots, t_n) \rfloor_{\mathcal{G}} = (t'_1, \dots, t'_n),$$

where $t'_1, \dots, t'_n$ are terms in $\mathcal{T}(\mathcal{G}, \mathcal{V})$ such that (a) $t'_i \sigma = t_i$ for all $1 \leq i \leq n$, (b) $\text{root}(x\sigma) \in \mathcal{F}$ for all $x \in \text{Dom}(\sigma)$, and (c) $\text{Dom}(\sigma) \cap \text{FV}(\phi) = \emptyset$ for some substitution σ such that $\text{Dom}(\sigma)$ is minimum in the sense of (b) and (c).

Notice that the operation $\lfloor \cdot \rfloor_{\mathcal{G}}$ is computable and the result is unique up to renaming of variables.

Example 3.5. Consider again $\mathcal{R}_{\text{isqrt}}$ in Example 3.3. We have that $\lfloor (\mathbf{s}(\mathbf{u}(i, n)), \mathbf{u}(i, n)) \rfloor_{\mathcal{G}_Z} = (\mathbf{s}(x), x)$.

The following proposition ensures that the operation can soundly be used for guaranteeing triviality of inductive theorems.

Proposition 3.6. Let $s \gtrsim t [\phi]$ be a constrained inequality and let $(s', t') = \lfloor (s, t) \rfloor_{\mathcal{G}}$. If $\phi \Rightarrow s' \gtrsim t'$ is $\mathcal{M}$ -valid, then $s \gtrsim t [\phi]$ is an inductive theorem of $\mathcal{R}$.

Proposition 3.6 is an immediate consequence of the following two lemmas.

Lemma 3.7. Let $s \gtrsim t [\phi]$ be a constrained inequality such that $s, t \in \mathcal{T}(\mathcal{G}, \mathcal{V})$. If $\phi \Rightarrow s \gtrsim t$ is $\mathcal{M}$ -valid, then $s \gtrsim t [\phi]$ is an inductive theorem of $\mathcal{R}$.

Proof. Assume that $\phi \Rightarrow s \gtrsim t$ is $\mathcal{M}$ -valid. Let σ be a substitution grounding s, t such that $\text{Ran}(\sigma|_{\text{FV}(\phi)}) \subseteq \mathcal{T}(\mathcal{G}, \mathcal{V})$ and $\phi\sigma$ is $\mathcal{M}$ -valid. Then it suffices to show that $\mathcal{M} \models s\sigma \downarrow_{\mathcal{R}} \gtrsim t\sigma \downarrow_{\mathcal{R}}$. Let $\sigma' = \{x \mapsto x\sigma \downarrow_{\mathcal{R}} \mid x \in \text{Dom}(\sigma) \cap \text{Var}(s, t)\}$. Note that $\text{Ran}(\sigma') \subseteq \mathcal{T}(\mathcal{G})$. By assumption, we have that $\mathcal{M} \models \phi\sigma' \Rightarrow s\sigma' \gtrsim t\sigma'$. Since $\mathcal{M} \models \phi\sigma$ implies $\mathcal{M} \models \phi\sigma'$, we have that $\mathcal{M} \models s\sigma' \gtrsim t\sigma'$, and hence $\mathcal{M} \models s\sigma' \downarrow_{\mathcal{R}} \gtrsim t\sigma' \downarrow_{\mathcal{R}}$. Since $s\sigma' \downarrow_{\mathcal{R}} \equiv s\sigma \downarrow_{\mathcal{R}}$ and $t\sigma' \downarrow_{\mathcal{R}} \equiv t\sigma \downarrow_{\mathcal{R}}$, we have that $\mathcal{M} \models s\sigma \downarrow_{\mathcal{R}} \gtrsim t\sigma \downarrow_{\mathcal{R}}$. $\square$

Lemma 3.8. Let $s \gtrsim t [\phi]$ be a constrained inequality and σ be a substitution such that $\text{Dom}(\sigma) \cap \text{FV}(\phi) = \emptyset$. If $s \gtrsim t [\phi]$ is an inductive theorem of $\mathcal{R}$, then so is $s\sigma \gtrsim t\sigma [\phi]$.

Proof. This lemma trivially follows from the definition of inductive theorems. $\square$

4. Inference rules for constrained inequalities

In this section, we present a framework for proving constrained inequalities to be inductive theorems. The framework performs implicit induction based on the RI principle and works on constrained TRSs that are ground convergent, sufficiently complete w.r.t. interpretable terms, and locally sound w.r.t. the associated structure. We suppose that the predicate $\gtrsim$ to form inequalities is transitive.

In the remainder of this section, we first describe basic ideas for adapting inference rules to our framework handling constrained inequalities (Section 4.1). Then, we define monotonicity of contexts and present a criterion for monotonicity (Section 4.2). After that, we formalize the framework (Section 4.3) and show some examples in which constrained inequalities are successfully proved to be inductive theorems within the framework (Section 4.4). Soundness of the framework will be proved in Section 5.

4.1. Ideas for adapting inference rules to inequalities

In principle, proof processes within RI frameworks proceed by application of three basic inference rules—Simplification, Deletion, and Expansion—to *proof states*. For simplicity, we here do not care about other inference rules such as the “Postulation” rule. A proof state has the form of a pair $(\mathcal{E}, \mathcal{H})$ of a set $\mathcal{E}$ of *conjectures* $s \approx t$ and a set $\mathcal{H}$ of *induction hypotheses* raised by the Expansion rule. Note that each induction hypothesis is an oriented equation such as $s \rightarrow t$ and can be applied as rewrite rules to conjectures. If we obtain a derivation starting with $(\mathcal{E}, \emptyset)$ and ending with $(\emptyset, \mathcal{H})$, then all the equations in $\mathcal{E}$ are proved to be inductive theorems of the given rewriting system. Since $\rightarrow_{\mathcal{R} \cup \mathcal{H}}$ is used as the measure of induction, termination of $\mathcal{R} \cup \mathcal{H}$ is necessary, and hence we orient induction hypotheses in a direction that preserves termination. The roles of the inference rules are briefly explained as follows:

- The Simplification rule rewrites a conjecture $C[l\sigma] \approx t \in \mathcal{E}$ into $C[r\sigma] \approx t$ with a rewrite rule or an induction hypothesis $l \rightarrow r \in \mathcal{R} \cup \mathcal{H}$.
- The Deletion rule removes from $\mathcal{E}$ a conjecture that is *clearly* an inductive theorem. Typically, this rule removes conjectures of the form $s \approx s$.
- The Expansion rule performs an exhaustive case analysis and splits a conjecture $s \approx t \in \mathcal{E}$ into multiple conjectures $s_1 \approx t, \dots, s_n \approx t$, so-called *subgoals* of $s \approx t$, such that if they are inductive theorems, then so is $s \approx t$. Moreover, this rule adds the original conjecture $s \approx t$ to $\mathcal{H}$ as an induction hypothesis with oriented like $s \rightarrow t$ so that $\mathcal{R} \cup \mathcal{H} \cup \{s \rightarrow t\}$ is terminating.

To directly handle inequalities within an RI framework, we change the elements of the set $\mathcal{E}$ from equations to inequalities. As for induction hypotheses, we must take into account not only the directions decided in the Expansion rule but also the original directions w.r.t. $\succsim$ unlike the case equations. For this reason, we modify proof states so that these two kinds of directions of induction hypotheses are taken into account.

Since both conjectures and induction hypotheses are now inequalities, positions to apply induction hypotheses are restricted. For example, let $s \succsim t$ be an induction hypothesis oriented from left to right and $C[s] \succsim u$ be a conjecture to be proved. Then, inferring a subgoal $C[t] \succsim u$ by applying the induction hypothesis is sound under the condition that $C[\]$ is “positively monotone”. On the other hand, to avoid incomplete inferences, we do not apply the same induction hypothesis to a conjecture $u \succsim C[s]$ to obtain a subgoal $u \succsim C[t]$.

For these reasons, our framework has the following changes from the existing ones for equations:

- The set $\mathcal{H}$ of induction hypotheses in each proof state $(\mathcal{E}, \mathcal{H})$ is split into $\mathcal{H}_{\rightarrow}$ and $\mathcal{H}_{\leftarrow}$ to fix the applicable directions. More precisely, a proof state in our framework is a triple $(\mathcal{E}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})$ of a set $\mathcal{E}$ of conjectures, a set $\mathcal{H}_{\rightarrow}$ of induction hypotheses applicable from left to right, and a set $\mathcal{H}_{\leftarrow}$ of induction hypotheses applicable from right to left.
- The Expansion rule is split into the L- and R-Expansion rules to distinguish which side of a conjecture is expanded. The L-Expansion (resp. R-Expansion) rule expands the left- (resp. right-) hand side of a conjecture and generates an induction hypothesis applicable from left to right (resp. from right to left), which is stored in $\mathcal{H}_{\rightarrow}$ (resp. $\mathcal{H}_{\leftarrow}$). Note that such a direction is independent of the meaning of the predicate symbol $\succsim$.
- Applicable positions of induction hypotheses are restricted to the inside of monotone contexts, whereas application of rewrite rules remains free from such a restriction. For this reason, the Simplification rule is split into the Rule-Application and L/R-Hypothesis-Application rules to distinguish between application of rewrite rules and that of induction hypotheses.

Soundness of induction hypothesis application in the above way relies on transitivity of the predicate symbol $\succsim$ that forms inequalities. Therefore, we require the predicate symbol $\succsim$ to be transitive. Here, transitivity of predicate symbols is defined as follows:

Definition 4.1. A binary predicate symbol p in $\mathcal{P}$ is said to be *transitive* under $\mathcal{M}$ if $p^{\mathcal{M}}$ is transitive, i.e., $p(x, y) \wedge p(y, z) \Rightarrow p(x, z)$ is $\mathcal{M}$ -valid, where x, y , and z are distinct variables.

Example 4.2. The predicate symbol $\succsim$ in Example 2.1 is transitive under $\mathcal{M}_{\mathbb{Z}}$.

Direct handling of inequalities enables us to utilize transitivity of binary predicates and to directly apply induction hypotheses in the form of inequalities. Thus, we can reduce a conjecture $u\sigma \gtrsim t[\phi]$ to a subgoal $v\sigma \gtrsim t[\phi]$ with an induction hypothesis $u \gtrsim v[\psi]$ such that $\phi \Rightarrow \psi\sigma$ is $\mathcal{M}$ -valid. As already seen in Section 1, such an application is often difficult in reducing inequalities to equations.

To adapt the Deletion rule to constrained inequalities, we have to adapt the notion of “trivial” conjectures. As already mentioned in Section 3.2, a constrained inequality $s \gtrsim t[\phi]$ is easily proved to be an inductive theorem of $\mathcal{R}$ if $\phi \Rightarrow s' \gtrsim t'$ is $\mathcal{M}$ -valid for $(s', t') = \lfloor s, t \rfloor_{\mathcal{G}}$. From this fact, the Deletion rule is modified to remove such trivial conjectures. If $\gtrsim$ is “reflexive”,⁶ then this new side condition covers the original one ($s \equiv t$ or ϕ is $\mathcal{M}$ -unsatisfiable).

4.2. Monotone contexts

Monotonicity of contexts used in this paper is defined by the built-in semantics and the rewrite relation as follows.

Definition 4.3. Let $\gtrsim : \tau \times \tau$ be a binary predicate symbol in $\mathcal{P}$ and ϕ be a formula over $(\mathcal{G}, \mathcal{P}, \mathcal{V})$. A context $C[\]$ is said to be *isotone* (resp. *antitone*) w.r.t. $\gtrsim$ and $\mathcal{R}$ under ϕ if $\mathcal{M} \models s \downarrow_{\mathcal{R}} \gtrsim t \downarrow_{\mathcal{R}}$ implies $\mathcal{M} \models (C\sigma[s]) \downarrow_{\mathcal{R}} \gtrsim (C\sigma[t]) \downarrow_{\mathcal{R}}$ (resp. $\mathcal{M} \models (C\sigma[t]) \downarrow_{\mathcal{R}} \gtrsim (C\sigma[s]) \downarrow_{\mathcal{R}}$) for every ground term s and t , and substitution σ grounding $C[\]$ such that $\phi\sigma$ is $\mathcal{M}$ -valid. If $\gtrsim$ and $\mathcal{R}$ are clear from the context, we simply say that $C[\]$ is isotone (resp. antitone) under ϕ .

Example 4.4. A context $\text{isqrt}(x) + \square$ is isotone w.r.t. $\gtrsim$ and $\mathcal{R}_{\text{isqrt}}$ under $\top$. A context $x \times \square$ is antitone w.r.t. $\gtrsim$ and $\mathcal{R}_{\text{isqrt}}$ under $x \not\approx 0$.

Related notions of context monotonicity can be found in the context of termination verification within dependency pair frameworks (e.g., [20, 14]). Note that monotonicity in their settings is defined by a quasi-order on terms constituting a reduction pair, and hence it is in general independent of the meanings of symbols under the built-in semantics unlike ours.

To test monotonicity of contexts, we can reuse the operation $\lfloor \cdot \rfloor_{\mathcal{G}}$ defined in Section 3.2. The following theorem presents a sufficient condition for monotonicity of contexts.

Theorem 4.5. Let $C[\]$ be a context, ϕ be a constraint, and x and y be fresh variables. Let $(s, t) = \lfloor (C[x], C[y]) \rfloor_{\mathcal{G}}$. Then, both of the following hold:

- (a) If $\phi \wedge x \gtrsim y \Rightarrow s \gtrsim t$ is $\mathcal{M}$ -valid, then $C[\]$ is isotone w.r.t. $\gtrsim$ and $\mathcal{R}$ under ϕ .
- (b) If $\phi \wedge x \gtrsim y \Rightarrow t \gtrsim s$ is $\mathcal{M}$ -valid, then $C[\]$ is antitone w.r.t. $\gtrsim$ and $\mathcal{R}$ under ϕ .

Proof. We here show only (a) since (b) can be proved analogously to (a). Assume that $\phi \wedge x \gtrsim y \Rightarrow s \gtrsim t$ is $\mathcal{M}$ -valid. Let u and v be ground terms and σ be a substitution grounding $C[\]$ such that $\phi\sigma$ is $\mathcal{M}$ -valid. We show that $\mathcal{M} \models u \downarrow_{\mathcal{R}} \gtrsim v \downarrow_{\mathcal{R}}$ implies $\mathcal{M} \models (C\sigma[u]) \downarrow_{\mathcal{R}} \gtrsim (C\sigma[v]) \downarrow_{\mathcal{R}}$. Assume that $\mathcal{M} \models u \downarrow_{\mathcal{R}} \gtrsim v \downarrow_{\mathcal{R}}$. Let $\theta = \{x \mapsto u \downarrow_{\mathcal{R}}, y \mapsto v \downarrow_{\mathcal{R}}\}$ and $(u', v') = \lfloor (C[u \downarrow_{\mathcal{R}}], C[v \downarrow_{\mathcal{R}}]) \rfloor_{\mathcal{G}}$. Since $(x \gtrsim y)\theta \equiv u \downarrow_{\mathcal{R}} \gtrsim v \downarrow_{\mathcal{R}}$, the formula $(x \gtrsim y)\theta$ is $\mathcal{M}$ -valid. Hence, by assumption, $\phi\theta \Rightarrow (s \gtrsim t)\theta$ is $\mathcal{M}$ -valid. Since $\phi\theta \equiv \phi$ and $(s \gtrsim t)\theta \equiv u' \gtrsim v'$, the formula $\phi \Rightarrow u' \gtrsim v'$ is $\mathcal{M}$ -valid. By Proposition 3.6 and the definition of inductive theorems, $\mathcal{M} \models (C[u \downarrow_{\mathcal{R}}])\sigma \downarrow_{\mathcal{R}} \gtrsim (C[v \downarrow_{\mathcal{R}}])\sigma \downarrow_{\mathcal{R}}$. Since $(C[u \downarrow_{\mathcal{R}}])\sigma \downarrow_{\mathcal{R}} \equiv (C\sigma[u]) \downarrow_{\mathcal{R}}$ and $(C[v \downarrow_{\mathcal{R}}])\sigma \downarrow_{\mathcal{R}} \equiv (C\sigma[v]) \downarrow_{\mathcal{R}}$, we have that $\mathcal{M} \models (C\sigma[u]) \downarrow_{\mathcal{R}} \gtrsim (C\sigma[v]) \downarrow_{\mathcal{R}}$. $\square$

Example 4.6. Let us consider the constrained TRS $\mathcal{R}_{\text{isqrt}}$ in Example 3.3 and a context $\text{isqrt}(z) + \square$. Given fresh variables x and y , we have that $\lfloor (\text{isqrt}(z)+x, \text{isqrt}(z)+y) \rfloor_{\mathcal{G}} = (v_{\text{isqrt}(z)}+x, v_{\text{isqrt}(z)}+y)$, and $x \gtrsim y \Rightarrow v_{\text{isqrt}(z)}+x \gtrsim v_{\text{isqrt}(z)}+y$ is $\mathcal{M}_{\mathbb{Z}}$ -valid. Therefore, it follows from Theorem 4.5 that $\text{isqrt}(z) + \square$ is isotone w.r.t. $\gtrsim$ and $\mathcal{R}_{\text{isqrt}}$ under $\top$.

4.3. Formalization of inference rules

Now we formalize the RI framework for constrained inequalities. A proof state of the framework is a triple $(\mathcal{E}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})$ of sets of constrained inequalities. Figure 1 illustrates a series of inference rules following the idea in

⁶We say that $\gtrsim$ is *reflexive* if $x \gtrsim y \Rightarrow y \gtrsim x$ is $\mathcal{M}$ -valid, where x and y are distinct variables.

Section 4.1: the notations $\vec{\mathcal{H}}$ and $\overleftarrow{\mathcal{H}}$ denote $\vec{\mathcal{H}} = \{s \rightarrow t[\phi] \mid s \gtrsim t[\phi] \in \mathcal{H}\}$ and $\overleftarrow{\mathcal{H}} = \{t \rightarrow s[\phi] \mid s \gtrsim t[\phi] \in \mathcal{H}\}$, respectively, and the sets $\text{Expd}_L(s \gtrsim t[\phi], p)$ and $\text{Expd}_R(s \gtrsim t[\phi], p)$ are defined as

$$\text{Expd}_L(C[s]_p \gtrsim t[\phi], p) = \{(C[r]_p)\sigma \gtrsim t\sigma \mid l \rightarrow r[\psi] \in \mathcal{R}, \sigma = \text{mgu}(s, l), \mathcal{R}\text{an}(\sigma|_{FV(\phi \wedge \psi)}) \subseteq \mathcal{T}(\mathcal{G}, \mathcal{V})\},$$

$$\text{Expd}_R(s \gtrsim C[t]_p[\phi], p) = \{s\sigma \gtrsim (C[r]_p)\sigma \mid l \rightarrow r[\psi] \in \mathcal{R}, \sigma = \text{mgu}(t, l), \mathcal{R}\text{an}(\sigma|_{FV(\phi \wedge \psi)}) \subseteq \mathcal{T}(\mathcal{G}, \mathcal{V})\}.$$

In the L- and R-Expansion rules, we consider existentially quantified variants $\tilde{\phi}$ of constraints ϕ so that the new induction hypothesis $s \gtrsim t[\tilde{\phi}]$ satisfies the variable condition of rewrite rules. Here, $\tilde{\phi}$ is formally defined to be $\exists x_1, \dots, x_n. \phi$ if the focused conjecture is $s \gtrsim t[\phi]$ and $FV(\phi) \setminus \text{Var}(s, t) = \{x_1, \dots, x_n\}$. Notice that this transformation of constraints does not change the meaning of induction hypotheses because constrained inequalities $s \gtrsim t[\phi]$ and $s \gtrsim t[\tilde{\phi}]$ can be informally read as “ $\phi \Rightarrow s \gtrsim t$ ” and “ $\tilde{\phi} \Rightarrow s \gtrsim t$ ”, respectively, and they are equivalent w.r.t. validity and satisfiability.

We write $(\mathcal{E}, \mathcal{H}_\rightarrow, \mathcal{H}_\leftarrow) \vdash_{RI} (\mathcal{E}', \mathcal{H}'_\rightarrow, \mathcal{H}'_\leftarrow)$ for a one-step deduction from $(\mathcal{E}, \mathcal{H}_\rightarrow, \mathcal{H}_\leftarrow)$ to $(\mathcal{E}', \mathcal{H}'_\rightarrow, \mathcal{H}'_\leftarrow)$ by an inference rule in Figure 1. The reflexive transitive closure of $\vdash_{RI}$ is denoted by $\vdash_{RI}^*$. To specify the applied inference rule, we may write $\vdash_{RuleApp}$, $\vdash_{LHypoApp}$, $\vdash_{RHypoApp}$, $\vdash_{Del}$, $\vdash_{LExp}$, and $\vdash_{RExp}$. We will show in Section 5 that if a derivation $(\mathcal{E}, \emptyset, \emptyset) \vdash_{RI}^* (\emptyset, \mathcal{H}_\rightarrow, \mathcal{H}_\leftarrow)$ on $\mathcal{R}$ is obtained, then all the constrained inequalities in $\mathcal{E}$ are inductive theorems of $\mathcal{R}$.

Satisfiability of constraints in the side conditions of the Rule-Application and L/R-Hypothesis-Application rules are intended to prevent meaningless application of the rules. Note that this condition is not necessary for soundness of the framework.

The side condition of the Deletion rule is improved from that of [33]. The Deletion rule in [33] removes conjectures of the form $s \gtrsim t[\phi]$ if (a) $s, t \in \mathcal{T}(\mathcal{G}, \mathcal{V})$ and $\phi \Rightarrow s \gtrsim t$ is $\mathcal{M}$ -valid or (b) ϕ is $\mathcal{M}$ -unsatisfiable. The improved Deletion rule enables us to remove conjectures with uninterpreted function symbols by introducing the operation $[\cdot]_{\mathcal{G}}$. Note that similar techniques of generalization for removal of conjectures have been proposed in the literature. The RI framework in [37], for example, allows us to remove a conjecture $s \approx t[\phi]$ such that $s \equiv s'\sigma$, $t \equiv t'\sigma$, $s', t' \in \mathcal{T}(\mathcal{G}, \mathcal{V})$, and $s' \approx t'$ is $\mathcal{M}$ -valid, provided that $\mathcal{R}$ is sufficiently complete w.r.t. interpretable terms. Compared to this, our improved Deletion rule is optimized in the sense that only a “minimum” substitution is considered.

The side conditions of our L- and R-Expansion rules include termination and reduction completeness, and therefore some criteria for these properties are required. We use termination provers instead of reduction orders as in [2], and methods for proving termination have already been mentioned in Section 2.4. To prove reduction completeness, we can use a straightforward adaptation of test-set-based criteria for ground reducibility [25]. Assume that a subsignature $C \subseteq \mathcal{F} \cup \mathcal{G}$ is given, where (a) $c : \tau \in C$ with $\tau \in \mathcal{S}_I$ implies $c \in \mathcal{G}$ and (b) $\mathcal{R}$ is quasi-reductive w.r.t. C in the sense defined in Section 2.4. Then, the root position of a term t is reduction-complete under a constraint ψ if $\psi\sigma \Rightarrow \bigvee_{\phi \in \text{Const}(t\sigma)} \phi$ is $\mathcal{M}$ -valid for all test-set-based substitutions⁷ σ with $\text{Dom}(\sigma) \supseteq \text{Var}(t)$, where Const is the same as the one in Section 2.4 and we consider a test set of the TRS $\{l \rightarrow r \mid l \rightarrow r[\phi] \in \mathcal{R}\}$ w.r.t. C .

4.4. Examples

We conclude this section by walking through some illustrative examples describing features of our framework. The first example explains how our framework works on the constrained TRS $\mathcal{R}_{\text{isqrt}}$ and its specifications in Example 3.3.

Example 4.7. Let us recall the constrained TRS $\mathcal{R}_{\text{isqrt}}$ in Example 3.3:

$$\mathcal{R}_{\text{isqrt}} = \left\{ \begin{array}{ll} \text{isqrt}(n) \rightarrow u(0, n), & \\ u(i, n) \rightarrow u(s(i), n) & [n \gtrsim s(i) \times s(i)], \\ u(i, n) \rightarrow i & [n \not\gtrsim s(i) \times s(i)] \end{array} \right\} \cup \mathcal{R}_{\text{arith}}.$$

As already seen in Example 2.2, $\mathcal{R}_{\text{isqrt}}$ is ground convergent, sufficiently complete w.r.t. interpretable terms, and locally sound w.r.t. $\mathcal{M}_{\mathbb{Z}}$. It is easy to see that the binary predicate symbol $\gtrsim$ is transitive under $\mathcal{M}_{\mathbb{Z}}$. As seen in Example 3.3, the following constrained inequalities represent the specification of the integer square root function:

$$\text{isqrt}(n) \gtrsim 0 \quad [n \gtrsim 0], \quad (8)$$

$$n \gtrsim \text{isqrt}(n) \times \text{isqrt}(n) \quad [n \gtrsim 0], \quad (9)$$

$$s(\text{isqrt}(n)) \times s(\text{isqrt}(n)) \gtrsim s(n) \quad [n \gtrsim 0]. \quad (10)$$

⁷A substitution σ is *test-set-based* if for each $x \in \text{Dom}(\sigma)$, $x\sigma$ is a renamed variant of a test set and $\text{Var}(x\sigma) \cap \text{Var}(y\sigma) = \emptyset$ for any $y \in \text{Dom}(\sigma)$ with $y \neq x$.

Rule-Application

$$\frac{(\mathcal{E} \uplus \{C[l\sigma] \gtrsim t[\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}{(\mathcal{E} \cup \{C[r\sigma] \gtrsim t[\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}$$

or

$$\frac{(\mathcal{E} \uplus \{s \gtrsim C[l\sigma][\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}{(\mathcal{E} \cup \{s \gtrsim C[r\sigma][\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}$$

where $l \rightarrow r[\psi] \in \mathcal{R}$, $\mathcal{R}an(\sigma|_{FV(\psi)}) \subseteq \mathcal{T}(\mathcal{G}, \mathcal{V})$, $\phi \Rightarrow \psi\sigma$ is $\mathcal{M}$ -valid, and ϕ is $\mathcal{M}$ -satisfiable.

L-Hypothesis-Application

$$\frac{(\mathcal{E} \uplus \{C[u\sigma] \gtrsim t[\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}{(\mathcal{E} \cup \{C[v\sigma] \gtrsim t[\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}$$

or

$$\frac{(\mathcal{E} \uplus \{s \gtrsim D[u\sigma][\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}{(\mathcal{E} \cup \{s \gtrsim D[v\sigma][\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}$$

where $u \gtrsim v[\psi] \in \mathcal{H}_{\rightarrow}$, $\mathcal{R}an(\sigma|_{FV(\psi)}) \subseteq \mathcal{T}(\mathcal{G}, \mathcal{V})$, $\phi \Rightarrow \psi\sigma$ is $\mathcal{M}$ -valid, ϕ is $\mathcal{M}$ -satisfiable, and $C[\]$ (resp. $D[\]$) is isotone (resp. antitone) under ϕ .

R-Hypothesis-Application

$$\frac{(\mathcal{E} \uplus \{s \gtrsim C[v\sigma][\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}{(\mathcal{E} \cup \{s \gtrsim C[u\sigma][\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}$$

or

$$\frac{(\mathcal{E} \uplus \{D[v\sigma] \gtrsim t[\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}{(\mathcal{E} \cup \{D[u\sigma] \gtrsim t[\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}$$

where $u \gtrsim v[\psi] \in \mathcal{H}_{\leftarrow}$, $\mathcal{R}an(\sigma|_{FV(\psi)}) \subseteq \mathcal{T}(\mathcal{G}, \mathcal{V})$, $\phi \Rightarrow \psi\sigma$ is $\mathcal{M}$ -valid, ϕ is $\mathcal{M}$ -satisfiable, and $C[\]$ (resp. $D[\]$) is isotone (resp. antitone) under ϕ .

Deletion

$$\frac{(\mathcal{E} \uplus \{s \gtrsim t[\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}{(\mathcal{E}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}$$

where, by letting $(s', t') = \lfloor (s, t) \rfloor_{\mathcal{G}}$, the formula $\phi \Rightarrow s' \gtrsim t'$ is $\mathcal{M}$ -valid.

L-Expansion

$$\frac{(\mathcal{E} \uplus \{s \gtrsim t[\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}{(\mathcal{E} \cup \text{Expd}_L(s \gtrsim t[\phi], p), \mathcal{H}_{\rightarrow} \cup \{s \gtrsim t[\tilde{\phi}]\}, \mathcal{H}_{\leftarrow})}$$

where $p \in \text{Pos}(s)$ is reduction-complete under ϕ , $\tilde{\phi}$ is an existentially quantified variant of ϕ such that $FV(\tilde{\phi}) = FV(\phi) \cap \mathcal{V}ar(s, t)$, $s \rightarrow t[\tilde{\phi}]$ is a rewrite rule, and $\mathcal{R} \cup \overrightarrow{\mathcal{H}_{\rightarrow}} \cup \overleftarrow{\mathcal{H}_{\leftarrow}} \cup \{s \rightarrow t[\tilde{\phi}]\}$ is terminating.

R-Expansion

$$\frac{(\mathcal{E} \uplus \{s \gtrsim t[\phi]\}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})}{(\mathcal{E} \cup \text{Expd}_R(s \gtrsim t[\phi], p), \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow} \cup \{s \gtrsim t[\tilde{\phi}]\})}$$

where $p \in \text{Pos}(t)$ is reduction-complete under ϕ , $\tilde{\phi}$ is an existentially quantified variant of ϕ such that $FV(\tilde{\phi}) = FV(\phi) \cap \mathcal{V}ar(s, t)$, $t \rightarrow s[\tilde{\phi}]$ is a rewrite rule, and $\mathcal{R} \cup \overrightarrow{\mathcal{H}_{\rightarrow}} \cup \overleftarrow{\mathcal{H}_{\leftarrow}} \cup \{t \rightarrow s[\tilde{\phi}]\}$ is terminating.

Figure 1. Inference rules of our RI framework for constrained inequalities.

Let us prove inductive validity of (8)–(10) under $\mathcal{R}_{\text{isqrt}}$ with our inference rules. Here, we need to add the following lemma inequalities to the set of conjectures, because divergence would occur without these as in the case of equations:

$$u(i, n) \gtrsim 0 \quad [i \gtrsim 0 \wedge n \gtrsim i \times i \wedge n \gtrsim 0], \quad (11)$$

$$n \gtrsim u(i, n) \times u(i, n) \quad [i \gtrsim 0 \wedge n \gtrsim i \times i \wedge n \gtrsim 0], \quad (12)$$

$$s(u(i, n)) \times s(u(i, n)) \gtrsim s(n) \quad [i \gtrsim 0 \wedge n \gtrsim i \times i \wedge n \gtrsim 0]. \quad (13)$$

Generation of these lemmas is out of scope of this paper and is left for future work. We start with the proof state $(\{(8)–(13)\}, \emptyset, \emptyset)$. First, we apply the L/R-Expansion rules to (11)–(13) as follows:

$$\left(\left(\begin{array}{ll} (8) & \text{isqrt}(n) \gtrsim 0 \\ (9) & n \gtrsim \text{isqrt}(n) \times \text{isqrt}(n) \\ (10) & s(\text{isqrt}(n)) \times s(\text{isqrt}(n)) \gtrsim s(n) \\ (11) & u(i, n) \gtrsim 0 \\ (12) & n \gtrsim u(i, n) \times u(i, n) \\ (13) & s(u(i, n)) \times s(u(i, n)) \gtrsim s(n) \end{array} \quad \begin{array}{l} [n \gtrsim 0], \\ [n \gtrsim 0], \\ [n \gtrsim 0], \\ [i \gtrsim 0 \wedge n \gtrsim i \times i \wedge n \gtrsim 0], \\ [i \gtrsim 0 \wedge n \gtrsim i \times i \wedge n \gtrsim 0], \\ [i \gtrsim 0 \wedge n \gtrsim i \times i \wedge n \gtrsim 0] \end{array} \right), \emptyset, \emptyset \right) \\ \vdash_{LExp} \vdash_{RExp} \vdash_{LExp} \\ \left(\begin{array}{ll} (8), (9), (10), & \\ (11a) & u(s(i), n) \gtrsim 0 \\ (11b) & i \gtrsim 0 \\ (12a) & n \gtrsim u(s(i), n) \times u(i, n) \\ (12b) & n \gtrsim i \times u(i, n) \\ (13a) & s(u(s(i), n)) \times s(u(i, n)) \gtrsim s(n) \\ (13b) & s(i) \times s(u(i, n)) \gtrsim s(n) \end{array} \quad \begin{array}{l} [\phi \wedge n \gtrsim s(i) \times s(i)], \\ [\phi \wedge n \not\gtrsim s(i) \times s(i)], \\ [\phi \wedge n \gtrsim s(i) \times s(i)], \\ [\phi \wedge n \not\gtrsim s(i) \times s(i)], \\ [\phi \wedge n \gtrsim s(i) \times s(i)], \\ [\phi \wedge n \not\gtrsim s(i) \times s(i)] \end{array} \right), \{(11), (13)\}, \{(12)\} \right)$$

where we let ϕ be $i \gtrsim 0 \wedge n \gtrsim i \times i \wedge n \gtrsim 0$. Here, the side conditions of the L/R-Expansion rules are satisfied. Actually, the conjecture (11) (resp. (12) and (13)) is expanded at the root position of the left-hand side (resp. position 1 of the right-hand side and 1.2 of the left-hand side), and the position is reduction-complete. Moreover, $\mathcal{R}_{\text{isqrt}} \cup \{u(i, n) \rightarrow 0 [\phi], n \gtrsim u(i, n) \times u(i, n) \rightarrow n [\phi], s(u(i, n)) \times s(u(i, n)) [\phi]\}$ is terminating. The inequality (11) (resp. (12) and (13)) is added to the set $\mathcal{H}_\rightarrow$ (resp. $\mathcal{H}_\leftarrow$ and $\mathcal{H}_\rightarrow$) and can be used as an induction hypothesis afterward. Now the obtained conjectures (12a)–(13b) can be reduced by the Rule-Application rule as follows:

$$\left(\{(8), (9), (10), (11a), (11b), (12a), (12b), (13a), (13b)\}, \{(11), (13)\}, \{(12)\} \right) \\ \vdash_{RuleApp}^* \\ \left(\begin{array}{ll} (8), (9), (10), (11a), (11b), & \\ (12a') & n \gtrsim u(s(i), n) \times u(s(i), n) \\ (12b') & n \gtrsim i \times i \\ (13a') & s(u(s(i), n)) \times s(u(s(i), n)) \gtrsim s(n) \\ (13b') & s(i) \times s(i) \gtrsim s(n) \end{array} \quad \begin{array}{l} [\phi \wedge n \gtrsim s(i) \times s(i)], \\ [\phi \wedge n \not\gtrsim s(i) \times s(i)], \\ [\phi \wedge n \gtrsim s(i) \times s(i)], \\ [\phi \wedge n \not\gtrsim s(i) \times s(i)] \end{array} \right), \{(11), (13)\}, \{(12)\} \right)$$

At this point, we can apply the Deletion rule to the conjectures (11a), (12a'), and (13a') since they are trivial inductive theorems:

$$\left(\{(8), (9), (10), (11a), (11b), (12a'), (12b'), (13a'), (13b')\}, \{(11), (13)\}, \{(12)\} \right) \\ \vdash_{Del}^* \\ \left(\{(8), (9), (10), (11a), (12a'), (13a')\}, \{(11), (13)\}, \{(12)\} \right).$$

The conjectures (11a), (12a'), and (13a') can be reduced by applying (11), (12), and (13) as induction hypotheses, which is done by the L/R-Hypothesis-Application rules. All the induction hypotheses are applied at the root position, i.e., they are applied inside the trivial context $\square$, which is isotone. The resulting conjectures (11a'), (12a''), and (13a'')

are trivial inductive theorems, and hence they can be deleted:

$$\begin{aligned} & (\{(8), (9), (10), (11a), (12a'), (13a')\}, \{(11), (13)\}, \{(12)\}) \\ & \vdash_{LHypoApp} \vdash_{RHypoApp} \vdash_{LHypoApp} \\ & \left(\left(\begin{array}{l} (8), (9), (10), \\ (11a') \quad 0 \gtrsim 0 \quad [\phi \wedge n \gtrsim s(i) \times s(i)], \\ (12a'') \quad n \gtrsim n \quad [\phi \wedge n \gtrsim s(i) \times s(i)], \\ (13a'') \quad s(n) \gtrsim s(n) \quad [\phi \wedge n \gtrsim s(i) \times s(i)] \end{array} \right), \{(11), (13)\}, \{(12)\} \right) \\ & \vdash_{Del}^* \\ & (\{(8), (9), (10)\}, \{(11), (13)\}, \{(12)\}). \end{aligned}$$

Up to this point, all the lemmas (11)–(13) have been proved. Now we get down to verification of the specifications (8)–(10) by using these lemmas stored in $\mathcal{H}_\rightarrow$ and $\mathcal{H}_\leftarrow$. By applying the Rule-Application rule, we can reduce the conjectures as follows:

$$\begin{aligned} & (\{(8), (9), (10)\}, \{(11), (13)\}, \{(12)\}) \\ & \vdash_{RuleApp}^* \\ & \left(\left(\begin{array}{l} (8') \quad u(0, n) \gtrsim 0 \\ (9') \quad n \gtrsim u(0, n) \times u(0, n) \\ (10') \quad s(u(0, n)) \times s(u(0, n)) \gtrsim 0 \end{array} \right), \begin{array}{l} [n \gtrsim 0], \\ [n \gtrsim 0], \\ [n \gtrsim 0], \end{array} \right), \{(11), (13)\}, \{(12)\} \right) \end{aligned}$$

The obtained conjectures (8')–(10') can be reduced by lemmas (11)–(13) by the L/R-Hypothesis-Application rules, and then we can delete all the conjectures:

$$\begin{aligned} & (\{(8'), (9'), (10')\}, \{(11), (13)\}, \{(12)\}) \\ & \vdash_{LHypoApp} \vdash_{RHypoApp} \vdash_{LHypoApp} \\ & \left(\left(\begin{array}{l} (8'') \quad 0 \gtrsim 0 \quad [n \gtrsim 0], \\ (9'') \quad n \gtrsim n \quad [n \gtrsim 0], \\ (10'') \quad s(n) \gtrsim s(n) \quad [n \gtrsim 0] \end{array} \right), \{(11), (13)\}, \{(12)\} \right) \\ & \vdash_{Del}^* \\ & (\emptyset, \{(11), (13)\}, \{(12)\}). \end{aligned}$$

Since the derivation $(\{(8)–(13)\}, \emptyset, \emptyset) \vdash_{RI}^* (\emptyset, \{(11), (13)\}, \{(12)\})$ is obtained, inequalities (8)–(13) are inductive theorems of $\mathcal{R}_{isqrt}$.

The next example illustrates a situation where the operation $\lfloor \cdot \rfloor_{\mathcal{G}}$ is essentially required to delete conjectures.

Example 4.8. Consider the constrained TRS $\mathcal{R}_{count}$ in Example 2.1 again:

$$\mathcal{R}_{count} = \left\{ \begin{array}{l} \text{count}(x, \text{nil}) \rightarrow 0, \\ \text{count}(x, \text{cons}(y, ys)) \rightarrow s(\text{count}(x, ys)) \quad [x \approx y], \\ \text{count}(x, \text{cons}(y, ys)) \rightarrow \text{count}(x, ys) \quad [x \not\approx y], \\ \text{length}(\text{nil}) \rightarrow 0, \\ \text{length}(\text{cons}(x, xs)) \rightarrow s(\text{length}(xs)) \end{array} \right\} \cup \mathcal{R}_{arith}.$$

Ground convergence, sufficient completeness w.r.t. interpretable terms, and local soundness w.r.t. $\mathcal{M}_{\mathbb{Z}}$ of $\mathcal{R}_{count}$ can be proved in the same way as $\mathcal{R}_{isqrt}$. Let us prove the following constrained inequality to be an inductive theorem of $\mathcal{R}_{count}$:

$$\text{length}(xs) \gtrsim \text{count}(x, xs). \quad (14)$$

We can construct the following derivation and obtain two conjectures (14b') and (14c').

$$\begin{aligned} & (\{(14) \text{ length}(xs) \gtrsim \text{count}(x, xs)\}, \emptyset, \emptyset) \\ & \vdash_{LExp} \\ & \left(\left(\begin{array}{l} (14a) \quad \text{length}(\text{nil}) \gtrsim 0, \\ (14b) \quad \text{length}(y :: ys) \gtrsim s(\text{count}(x, ys)) \quad [x \approx y], \\ (14c) \quad \text{length}(y :: ys) \gtrsim \text{count}(x, ys) \quad [x \not\approx y] \end{array} \right), \{(14)\}, \emptyset \right) \end{aligned}$$

$$\begin{array}{l}
\vdash_{RuleApp}^* \vdash_{HypoApp}^* \\
\left(\left\{ \begin{array}{l} (14a') \quad 0 \gtrsim 0, \\ (14b') \quad s(\text{length}(ys)) \gtrsim s(\text{length}(ys)) \quad [x \approx y], \\ (14c') \quad s(\text{length}(ys)) \gtrsim \text{length}(ys) \quad [x \not\approx y] \end{array} \right\}, \{(14)\}, \emptyset \right) \\
\vdash_{Del} \\
\left(\left\{ \begin{array}{l} (14b') \quad s(\text{length}(ys)) \gtrsim s(\text{length}(ys)) \quad [x \approx y], \\ (14c') \quad s(\text{length}(ys)) \gtrsim \text{length}(ys) \quad [x \not\approx y] \end{array} \right\}, \{(14)\}, \emptyset \right).
\end{array}$$

Although the conjectures (14b') and (14c') have uninterpretable symbols, it is easy to see that they are inductive theorems of $\mathcal{R}_{\text{count}}$. To remove these conjectures, we can use the operation $\lfloor \cdot \rfloor_{\mathcal{G}}$ to replace each uninterpretable subterm of the conjectures with a fresh variable. In our example, to remove (14c'), we apply the operation as follows:

$$\lfloor s(\text{length}(ys)), \text{length}(ys) \rfloor_{\mathcal{G}} = (s(xs), xs).$$

Then, the obtained formula $x \not\approx y \Rightarrow s(xs) \gtrsim xs$ is $\mathcal{M}_{\mathbb{Z}}$ -valid, which implies that (14c') is an inductive theorem of $\mathcal{R}_{\text{count}}$, and hence (14c') can be removed by our Deletion rule. In the same way, we can also remove the conjecture (14b') as follows:

$$(\{(14b'), (14c')\}, \{(14)\}, \emptyset) \vdash_{Del}^* (\emptyset, \{(14)\}, \emptyset).$$

In the following example, we recall $\mathcal{R}_{\text{pow}}$ in Section 1 and look at how non-trivial contexts are handled in application of induction hypotheses.

Example 4.9. Let us consider again the constrained TRS $\mathcal{R}_{\text{pow}}$ over $(\emptyset, \{int\}, \{\text{pow} : int \times int \rightarrow int\}, \mathcal{G}_{\mathbb{Z}}, \mathcal{P}_{\mathbb{Z}}, \mathcal{M}_{\mathbb{Z}})$ in Section 1:

$$\mathcal{R}_{\text{pow}} = \left\{ \begin{array}{ll} \text{pow}(x, y) \rightarrow s(0) & [y \not\gtrsim s(0)], \\ \text{pow}(x, y) \rightarrow x \times \text{pow}(x, p(y)) & [y \gtrsim s(0)] \end{array} \right\} \cup \mathcal{R}_{\text{arith}}.$$

Let us prove the following constrained inequality to be an inductive theorem of $\mathcal{R}_{\text{pow}}$:

$$\text{pow}(s^2(0), n) \gtrsim s(n) \quad [n \gtrsim 0]. \quad (15)$$

By applying the L-Expansion and Deletion rules, we obtain the following derivation:

$$\begin{array}{l}
(\{(15) \text{ pow}(s^2(0), n) \gtrsim s(n) \quad [n \gtrsim 0]\}, \emptyset, \emptyset) \\
\vdash_{LExp} \\
\left(\left\{ \begin{array}{l} (15a) \quad s(0) \gtrsim s(n) \quad [n \gtrsim 0 \wedge n \not\gtrsim s(0)], \\ (15b) \quad s^2(0) \times \text{pow}(s^2(0), p(n)) \gtrsim s(n) \quad [n \gtrsim 0 \wedge n \gtrsim s(0)] \end{array} \right\}, \{(15)\}, \emptyset \right) \\
\vdash_{Del} \\
(\{(15b) \quad s^2(0) \times \text{pow}(s^2(0), p(n)) \gtrsim s(n) \quad [s(n) \gtrsim 0 \wedge n \gtrsim s(0)]\}, \{(15)\}, \emptyset).
\end{array}$$

In this example, if we replace $\text{pow}(s^2(0), n)$ in the left-hand side of (15b) with $s(n)$ by the induction hypothesis (15), the verification will be completed. As described in Section 1, this replacement is sound because $x \gtrsim y \Rightarrow s^2(0) \times x \gtrsim s^2(0) \times y$ is $\mathcal{M}_{\mathbb{Z}}$ -valid, i.e., the context $s^2(0) \times \square$ is isotone. Our L/R-Hypothesis-Application rules enable such a replacement, and we can now complete the verification of (15) as follows:

$$\begin{array}{l}
(\{(15b)\}, \{(15)\}, \emptyset) \\
\vdash_{LHypoApp} \\
(\{(15b') \quad s^2(0) \times s(p(n)) \gtrsim s(n) \quad [s(n) \gtrsim 0 \wedge n \gtrsim s(0)]\}, \{(15)\}, \emptyset) \\
\vdash_{Del} \\
(\emptyset, \{(15)\}, \emptyset).
\end{array}$$

The obtained derivation $(\{(15)\}, \emptyset, \emptyset) \vdash_{RI}^* (\emptyset, \{(15)\}, \emptyset)$ implies that (15) is an inductive theorem of $\mathcal{R}_{\text{pow}}$.

Our RI framework is not *refutationally sound*, i.e., one cannot disprove the initial conjectures even if an invalid inequality is derived. The following example illustrates such a situation.

Example 4.10. The following is a derivation on the constrained TRS $\mathcal{R}_{\text{pow}}$ starting with a conjecture $\text{pow}(s^2(0), n) \gtrsim n [n \gtrsim 0]$:

$$\begin{aligned}
 & (\{(16) \text{pow}(s^2(0), n) \gtrsim n [n \gtrsim 0]\}, \emptyset, \emptyset) \\
 & \quad \vdash_{LExp} \left(\left\{ \begin{array}{l} (16a) \quad s(0) \gtrsim n [n \gtrsim 0 \wedge n \not\gtrsim s(0)], \\ (16b) \quad s^2(0) \times \text{pow}(s^2(0), p(n)) \gtrsim n [n \gtrsim 0 \wedge n \gtrsim s(0)] \end{array} \right\}, \{(16)\}, \emptyset \right) \\
 & \quad \vdash_{Del} (\{(16b) s^2(0) \times \text{pow}(s^2(0), p(n)) \gtrsim n [n \gtrsim 0 \wedge n \gtrsim s(0)]\}, \{(16)\}, \emptyset) \\
 & \quad \vdash_{LHypoApp} (\{(16b') s^2(0) \times p(n) \gtrsim s(n) [s(n) \gtrsim 0 \wedge n \gtrsim s(0)]\}, \{(16)\}, \emptyset).
 \end{aligned}$$

The original conjecture (16) is an inductive theorem of $\mathcal{R}_{\text{pow}}$, but the derived subgoal (16b') is not. This non-refutational-soundness is due to the fact that application of the L/R-Hypothesis-Application rules may cause approximation of conjectures, i.e., the L/R-Hypothesis-Application rules are not complete in general. In some cases, incomplete inferences can be avoided by adjusting the timing of application of the L/R-Hypothesis-Application rules.

5. Soundness of the framework

In this section, we prove soundness of our framework proposed in Section 4. In Section 5.1, we present an abstract principle of RI for our framework. In Section 5.2, we define and discuss some binary relations over ground terms that are induced by given constrained inequalities. Section 5.3 shows some properties of monotone contexts. Finally, in Section 5.4, we prove the soundness theorem. Recall that a given constrained TRS $\mathcal{R}$ is assumed to be ground convergent, sufficiently complete w.r.t. interpretable terms, and locally sound w.r.t. $\mathcal{M}$ unless noted otherwise.

5.1. Abstract principles of RI

In [27, 3], abstract principles of RI have been presented in terms of *abstract reduction systems* [5]. Koike and Toyama first revealed in [27] that an abstract principle of RI is presented as the following proposition.

Proposition 5.1 ([27, Lemma 3.4]). *Let $\rightarrow_1$ and $\rightarrow_2$ be binary relations on A such that*

- $\rightarrow_1 \subseteq \rightarrow_2$,
- $\rightarrow_2$ is well-founded, and
- $\rightarrow_2 \subseteq \rightarrow_1 \circ \rightarrow_2^* \circ \leftarrow_2^*$.

Then, $\leftarrow_1^ = \leftarrow_2^*$.*

Although several generalized variants of the principle are presented in, e.g., [1, 3, 37], they do not fit our setting. This is because we essentially need a more general abstract principle described with four relations to state correctness of our framework. Hence, we adapt Proposition 5.1 to our setting in the following theorem.

Theorem 5.2. *Let $\rightarrow_1, \rightarrow_2, \rightarrow_3$, and $\rightarrow_4$ be binary relations on A such that*

- $\rightarrow_1 \subseteq \rightarrow_2$,
- $\rightarrow_1 \subseteq \rightarrow_3$,
- $\rightarrow_2 \cup \rightarrow_3$ is well-founded,
- $\rightarrow_2 \subseteq \rightarrow_1 \circ \rightarrow_2^* \circ (\leftarrow_1^* \cup \rightarrow_4) \circ \leftarrow_3^*$, and
- $\leftarrow_3 \subseteq \rightarrow_2^* \circ (\leftarrow_1^* \cup \rightarrow_4) \circ \leftarrow_3^* \circ \leftarrow_1$.

Then, $\rightarrow_2^ \subseteq (\leftarrow_1 \cup \rightarrow_4)^*$ and $\leftarrow_3^* \subseteq (\leftarrow_1 \cup \rightarrow_4)^*$.*

Proof. We prove the following two statements simultaneously by Noetherian induction on $\rightarrow_2 \cup \rightarrow_3$:

(i) If $x \rightarrow_2^* y$, then $x (\leftrightarrow_1 \cup \rightarrow_4)^* y$.

(ii) If $x \rightarrow_3^* y$, then $x (\leftrightarrow_1 \cup \leftarrow_4)^* y$.

The base case is obvious since $x = y$. For the induction case, suppose that $x (\rightarrow_2 \cup \rightarrow_3) z (\rightarrow_2 \cup \rightarrow_3)^* y$. We consider the following cases:

- Suppose that $x \rightarrow_2 z \rightarrow_2^* y$. Then, we have that $x \rightarrow_1 u \rightarrow_2^* v (\leftrightarrow_1^* \cup \rightarrow_4) w \leftarrow_3^* z$ for some $u, v, w \in A$. Since $x (\rightarrow_2 \cup \rightarrow_3) z \rightarrow_2^* y$, $x (\rightarrow_2 \cup \rightarrow_3) z \rightarrow_3^* w$, and $x (\rightarrow_2 \cup \rightarrow_3) u \rightarrow_2^* v$, it follows from the induction hypotheses that $z (\leftrightarrow_1 \cup \rightarrow_4)^* y$, $z (\leftrightarrow_1 \cup \leftarrow_4)^* w$, and $u (\leftrightarrow_1 \cup \rightarrow_4)^* v$. Hence, $x (\leftrightarrow_1 \cup \rightarrow_4)^* y$, implying (i).
- Suppose that $x \rightarrow_3 z \rightarrow_3^* y$. Then, we have that $x \rightarrow_1 u \rightarrow_3^* v (\leftrightarrow_1^* \cup \leftarrow_4) w \leftarrow_2^* z$ for some $u, v, w \in A$. Since $x (\rightarrow_2 \cup \rightarrow_3) z \rightarrow_3^* y$, $x (\rightarrow_2 \cup \rightarrow_3) z \rightarrow_2^* w$, and $x (\rightarrow_2 \cup \rightarrow_3) u \rightarrow_3^* v$, it follows from the induction hypotheses that $z (\leftrightarrow_1 \cup \leftarrow_4)^* y$, $z (\leftrightarrow_1 \cup \rightarrow_4)^* w$, and $u (\leftrightarrow_1 \cup \leftarrow_4)^* v$. Hence, $x (\leftrightarrow_1 \cup \leftarrow_4)^* y$, implying (ii).

Therefore, both (i) and (ii) hold when $x (\rightarrow_2 \cup \rightarrow_3) z (\rightarrow_2 \cup \rightarrow_3)^* y$. $\square$

5.2. Binary relations induced by constrained inequalities

We define four relations over ground terms that are determined by a given set $\mathcal{E}$ of constrained inequalities.

Definition 5.3. Let $\succsim : \tau \times \tau$ be a binary predicate symbol in $\mathcal{P}$, $\mathcal{E}$ be a set of inequalities. We define the relations $\succsim_{\mathcal{E}}^{iso}$, $\succsim_{\mathcal{E}}^{anti}$, and $\succsim_{\mathcal{E}}^{mono}$ over ground terms as follows:

$$\begin{aligned} \succsim_{\mathcal{E}}^{\square} &= \{(s\sigma, t\sigma) \mid s \succsim t[\phi] \in \mathcal{E}, \sigma \text{ grounds } s \text{ and } t, \text{Ran}(\sigma|_{FV(\phi)}) \subseteq (\mathcal{T}(\mathcal{G}, \mathcal{V})), \phi\sigma \text{ is } \mathcal{M}\text{-valid}\}, \\ \succsim_{\mathcal{E}}^{iso} &= \{(C[s], C[t]) \mid s \succsim_{\mathcal{E}}^{\square} t, C[\] \in \mathcal{T}_{\square}(\mathcal{F} \cup \mathcal{G}, \emptyset) \text{ is isotone under } \top\}, \\ \succsim_{\mathcal{E}}^{anti} &= \{(C[s], C[t]) \mid s \succsim_{\mathcal{E}}^{\square} t, C[\] \in \mathcal{T}_{\square}(\mathcal{F} \cup \mathcal{G}, \emptyset) \text{ is antitone under } \top\}, \\ \succsim_{\mathcal{E}}^{mono} &= \succsim_{\mathcal{E}}^{iso} \cup \succsim_{\mathcal{E}}^{anti}. \end{aligned}$$

The relations imitate the rewrite relation $\rightarrow_{\mathcal{R}}$. Superscripts indicate what kind of contexts is considered there. For instance, the superscript $\square$ means that only the trivial context is considered.

Next, we define another relation $\succsim_{\mathcal{R}}$ over ground terms, which is determined by a given constrained TRS $\mathcal{R}$ over and the interpretation of $\succsim$ under $\mathcal{M}$.

Definition 5.4. Let $\succsim : \tau \times \tau$ be a binary predicate symbol in $\mathcal{P}$. We define the relation $\succsim_{\mathcal{R}}$ over ground terms as follows:

$$\succsim_{\mathcal{R}} = \{(s, t) \mid s, t \in \mathcal{T}(\mathcal{F} \cup \mathcal{G}), s, t : \tau, \mathcal{M} \models s \downarrow_{\mathcal{R}} \succsim t \downarrow_{\mathcal{R}}\}.$$

Notice that we have that $s \downarrow_{\mathcal{R}}, t \downarrow_{\mathcal{R}} \in \mathcal{T}(\mathcal{G})$ in this definition.

The following lemma states that the inclusion $\succsim_{\mathcal{E}}^{mono} \subseteq \succsim_{\mathcal{R}}$ is a necessary and sufficient condition for all inequalities in $\mathcal{E}$ to be inductive theorems of $\mathcal{R}$.

Lemma 5.5. Let $\mathcal{E}$ be a set of constrained inequalities. Every constrained inequality in $\mathcal{E}$ is an inductive theorem of $\mathcal{R}$ iff $\succsim_{\mathcal{E}}^{mono} \subseteq \succsim_{\mathcal{R}}$.

Proof. We first prove the *if* part. Let $s \succsim t[\phi]$ be an inequality in $\mathcal{E}$ and σ be a substitution grounding s and t such that $\mathcal{M} \models \phi\sigma$. Since $s \succsim t[\phi] \in \mathcal{E}$, it follows from the definition of $\succsim_{\mathcal{E}}^{mono}$ that $s\sigma \succsim_{\mathcal{E}}^{mono} t\sigma$. By assumption, we have that $s\sigma \succsim_{\mathcal{R}} t\sigma$. By definition, we have that $\mathcal{M} \models s\sigma \downarrow_{\mathcal{R}} \succsim t\sigma \downarrow_{\mathcal{R}}$. Therefore, $s \succsim t[\phi]$ is an inductive theorem of $\mathcal{R}$.

Next, we prove the *only-if* part. Let $s \succsim_{\mathcal{E}}^{mono} t$. We assume w.l.o.g. that $s \succsim_{\mathcal{E}}^{iso} t$ since the other case $s \succsim_{\mathcal{E}}^{anti} t$ can be proved analogously. Then, by definition, there exist a constrained inequality $s' \succsim t'[\phi] \in \mathcal{E}$, an isotone context $C[\]$, and a ground substitution σ such that $s \equiv C[s'\sigma]$, $t \equiv C[t'\sigma]$, $\text{Ran}(\sigma|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$, and $\mathcal{M} \models \phi\sigma$. Since $s' \succsim t'[\phi]$ is an inductive theorem of $\mathcal{R}$ by assumption, we have that $\mathcal{M} \models s'\sigma \downarrow_{\mathcal{R}} \succsim t'\sigma \downarrow_{\mathcal{R}}$. By the definition of isotone contexts, $\mathcal{M} \models C[s'\sigma] \downarrow_{\mathcal{R}} \succsim C[t'\sigma] \downarrow_{\mathcal{R}}$, i.e., $\mathcal{M} \models s \downarrow_{\mathcal{R}} \succsim t \downarrow_{\mathcal{R}}$. Therefore, $s \succsim_{\mathcal{R}} t$. $\square$

Transitive predicate symbols have the following property regarding $\succsim_{\mathcal{R}}$.

Lemma 5.6. *If $\succsim$ is transitive under $\mathcal{M}$, then $(\leftrightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{R}} \circ \leftrightarrow_{\mathcal{R}}^*)^+ = \succsim_{\mathcal{R}}$ on ground terms.*

Proof. By definition, we have that $\leftrightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{R}} \circ \leftrightarrow_{\mathcal{R}}^* = \succsim_{\mathcal{R}}$. Since $\succsim$ is transitive under $\mathcal{M}$, $\succsim_{\mathcal{R}}$ is also transitive, and hence $\succsim_{\mathcal{R}}^+ = \succsim_{\mathcal{R}}$. Therefore, $(\leftrightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{R}} \circ \leftrightarrow_{\mathcal{R}}^*)^+ = \succsim_{\mathcal{R}}^+ = \succsim_{\mathcal{R}}$. $\square$

The following proposition describes some basic relations among $\succsim_{\mathcal{E}}^{\square}$, $\succsim_{\mathcal{E}}^{iso}$, and $\succsim_{\mathcal{E}}^{anti}$.

Proposition 5.7. *Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be sets of constrained inequalities. If $\succsim_{\mathcal{E}_1}^{\square} \subseteq \succsim_{\mathcal{E}_2}^{\square}$, then $\succsim_{\mathcal{E}_1}^{iso} \subseteq \succsim_{\mathcal{E}_2}^{iso}$ and $\succsim_{\mathcal{E}_1}^{anti} \subseteq \succsim_{\mathcal{E}_2}^{anti}$.*

Using Proposition 5.7 and the definitions of $\succsim_{\mathcal{E}}^{iso}$ and $\succsim_{\mathcal{E}}^{anti}$, we can immediately derive the following lemma.

Lemma 5.8. *Let $\mathcal{E}$, $\mathcal{E}_1$, and $\mathcal{E}_2$ be sets of constrained inequalities, $\mathcal{R}$ be a constrained TRS, and let $C[\]$ and $D[\]$ be contexts without variables. If $C[\]$ (resp. $D[\]$) is isotone (resp. antitone) under $\top$, then all of the following statements hold:*

- (a) *If $s \rightarrow_{\mathcal{R}} \circ \succsim_{\mathcal{E}}^{\square} t$, then $C[s] \rightarrow_{\mathcal{R}} \circ \succsim_{\mathcal{E}}^{iso} C[t]$ and $D[t] \rightarrow_{\mathcal{R}} \circ \succsim_{\mathcal{E}}^{anti} D[s]$.*
- (b) *If $s \succsim_{\mathcal{E}}^{\square} \circ \leftarrow_{\mathcal{R}} t$, then $C[s] \succsim_{\mathcal{E}}^{iso} \circ \leftarrow_{\mathcal{R}} C[t]$ and $D[t] \succsim_{\mathcal{E}}^{anti} \circ \leftarrow_{\mathcal{R}} D[s]$.*
- (c) *If $\succsim_{\mathcal{E}_1}^{\square} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \succsim_{\mathcal{E}_2}^{\square} \circ \leftarrow_{\mathcal{R}}^*$, then $\succsim_{\mathcal{E}_1}^{iso} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \succsim_{\mathcal{E}_2}^{iso} \circ \leftarrow_{\mathcal{R}}^*$ and $\succsim_{\mathcal{E}_1}^{anti} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}_2}^{anti} \circ \leftarrow_{\mathcal{R}}^+$.*
- (d) *If $\succsim_{\mathcal{E}_1}^{\square} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}_2}^{\square} \circ \leftarrow_{\mathcal{R}}^+$, then $\succsim_{\mathcal{E}_1}^{iso} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}_2}^{iso} \circ \leftarrow_{\mathcal{R}}^+$ and $\succsim_{\mathcal{E}_1}^{anti} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \succsim_{\mathcal{E}_2}^{anti} \circ \leftarrow_{\mathcal{R}}^*$.*

Proof. Claims (a) and (b) follow from the definitions of $\succsim_{\mathcal{E}}^{\square}$, $\succsim_{\mathcal{E}}^{iso}$, and $\succsim_{\mathcal{E}}^{anti}$. Claims (c) and (d) follow from Proposition 5.7. $\square$

5.3. Properties of monotone contexts

The following lemma states that compositions of monotone contexts are also monotone.

Lemma 5.9. *Let $C[\]$ and $D[\]$ be contexts. Then, all of the following statements hold:*

- (a) *If $C[\]$ is isotone under ϕ and $D[\]$ is isotone under ϕ , then $D[C[\]]$ is isotone under ϕ .*
- (b) *If $C[\]$ is isotone under ϕ and $D[\]$ is antitone under ϕ , then $D[C[\]]$ is antitone under ϕ .*
- (c) *If $C[\]$ is antitone under ϕ and $D[\]$ is isotone under ϕ , then $D[C[\]]$ is antitone under ϕ .*
- (d) *If $C[\]$ is antitone under ϕ and $D[\]$ is antitone under ϕ , then $D[C[\]]$ is isotone under ϕ .*

Proof. We here show only (a) since the others can be proved analogously to (a). Let $C[\]$ and $D[\]$ be contexts isotone under ϕ . Let s and t be terms, and σ be a substitution grounding $C[\]$ and $D[\]$ such that $\phi\sigma$ is $\mathcal{M}$ -valid. Since $D[\]$ is isotone under ϕ and $\phi\sigma$ is $\mathcal{M}$ -valid, we have that $\mathcal{M} \models (C\sigma[s])\downarrow_{\mathcal{R}} \succsim (C\sigma[t])\downarrow_{\mathcal{R}} \Rightarrow (D\sigma[C\sigma[s]])\downarrow_{\mathcal{R}} \succsim (D\sigma[C\sigma[t]])\downarrow_{\mathcal{R}}$. Since $C[\]$ is isotone under ϕ and $\phi\sigma$ is $\mathcal{M}$ -valid, we have that $\mathcal{M} \models s\downarrow_{\mathcal{R}} \succsim t\downarrow_{\mathcal{R}} \Rightarrow (C\sigma[s])\downarrow_{\mathcal{R}} \succsim (C\sigma[t])\downarrow_{\mathcal{R}}$, and hence $\mathcal{M} \models s\downarrow_{\mathcal{R}} \succsim t\downarrow_{\mathcal{R}} \Rightarrow (D\sigma[C\sigma[s]])\downarrow_{\mathcal{R}} \succsim (D\sigma[C\sigma[t]])\downarrow_{\mathcal{R}}$. Therefore, $D[C[\]]$ is isotone under ϕ . $\square$

Monotonicity of contexts is preserved under substitutions.

Lemma 5.10. *Let $C[\]$ be a context, ϕ be a constraint, and σ be a substitution grounding $C[\]$ such that $\phi\sigma$ is $\mathcal{M}$ -valid. If $C[\]$ is isotone (resp. antitone) under ϕ , then $C\sigma[\]$ is isotone (resp. antitone) under $\top$.*

Proof. Suppose that $C[\]$ is isotone (resp. antitone) under ϕ , and let s and t be terms. Since $\phi\sigma$ is $\mathcal{M}$ -valid, it follows by definition that $\mathcal{M} \models s\downarrow_{\mathcal{R}} \succsim t\downarrow_{\mathcal{R}} \Rightarrow (C\sigma[s])\downarrow_{\mathcal{R}} \succsim (C\sigma[t])\downarrow_{\mathcal{R}}$ (resp. $\mathcal{M} \models s\downarrow_{\mathcal{R}} \succsim t\downarrow_{\mathcal{R}} \Rightarrow (C\sigma[t])\downarrow_{\mathcal{R}} \succsim (C\sigma[s])\downarrow_{\mathcal{R}}$). Since σ grounds $C[\]$, we have that, for any substitution θ , $\mathcal{M} \models s\downarrow_{\mathcal{R}} \succsim t\downarrow_{\mathcal{R}} \Rightarrow (C\sigma\theta[s])\downarrow_{\mathcal{R}} \succsim (C\sigma\theta[t])\downarrow_{\mathcal{R}}$ (resp. $\mathcal{M} \models s\downarrow_{\mathcal{R}} \succsim t\downarrow_{\mathcal{R}} \Rightarrow (C\sigma\theta[t])\downarrow_{\mathcal{R}} \succsim (C\sigma\theta[s])\downarrow_{\mathcal{R}}$). Hence, $C\sigma[\]$ is isotone (resp. antitone) under $\top$. $\square$

5.4. Main theorem

In the remainder of this section, we show the soundness theorem of our framework. The entire proof basically follows the way of [37].

The following lemma is a slight modification of [37, Lemma A.1.3].

Lemma 5.11. *Suppose that $\mathcal{R}$ is terminating and locally sound w.r.t. $\mathcal{M}$. Let s and t be terms, ϕ be a constraint, and σ be a substitution grounding s and t such that $\mathcal{Ran}(\sigma|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$. If $\mathcal{M} \models \phi\sigma$, then there exists a normalized substitution θ grounding s and t such that $s\sigma \rightarrow_{\mathcal{R}}^* s\theta$, $t\sigma \rightarrow_{\mathcal{R}}^* t\theta$, $\mathcal{Ran}(\theta|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$, and $\mathcal{M} \models \phi\theta$.*

Proof. Suppose that $\mathcal{M} \models \phi\sigma$, and let $\theta = \{x\sigma \downarrow_{\mathcal{R}} \mid x \in \text{Dom}(\sigma)\}$. Then, it is trivial that $s\sigma \rightarrow_{\mathcal{R}}^* s\theta$ and $t\sigma \rightarrow_{\mathcal{R}}^* t\theta$. Local soundness w.r.t. $\mathcal{M}$ ensures that $\mathcal{Ran}(\theta|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$ and $\mathcal{M} \models \phi\theta$. $\square$

The next lemma explains existence of a ground substitution with certain properties corresponding to a given ground substitution.

Lemma 5.12. *Let s and t be terms, ϕ be a constraint, and σ be a substitution grounding s and t such that $FV(\phi) \subseteq \text{Var}(s, t)$. Let $X = \{x \in \text{Dom}(\sigma) \mid x\sigma \in \mathcal{T}(\mathcal{G}, \mathcal{V})\}$. If $\phi(\sigma|_X)$ is $\mathcal{M}$ -valid, then there exists a substitution θ grounding s and t such that $s\sigma \rightarrow_{\mathcal{R}}^* s\theta$, $t\sigma \rightarrow_{\mathcal{R}}^* t\theta$, $\mathcal{M} \models \phi\theta$, and $\sigma|_X = \theta|_X$.*

Proof. Suppose that $\phi(\sigma|_X)$ is $\mathcal{M}$ -valid. Let $\theta = \sigma|_X \cup \{x \mapsto x\sigma \downarrow_{\mathcal{R}} \mid x \in \text{Dom}(\sigma) \setminus X\}$. It is easy to see that $s\sigma \rightarrow_{\mathcal{R}}^* s\theta$, $t\sigma \rightarrow_{\mathcal{R}}^* t\theta$, and $\sigma|_X = \theta|_X$. By ground convergence and sufficient completeness, we have that $\mathcal{Ran}(\theta) \subseteq \mathcal{T}(\mathcal{G})$, and hence $\mathcal{M} \models \phi\theta$. $\square$

The following lemma describes the basic properties of the L/R-Expansion rules.

Lemma 5.13. *Let $s \gtrsim t[\phi]$ be a constrained inequality and σ be a normalized substitution grounding s and t such that $\mathcal{Ran}(\sigma|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$ and $\mathcal{M} \models \phi\sigma$. Then, both of the following statements hold:*

- (a) *If a position p of s is reduction-complete under ϕ , then $s\sigma \rightarrow_{\mathcal{R}} \circ \gtrsim_{\text{Expd}_L(s \gtrsim t[\phi], p)}^{\square} t\sigma$.*
- (b) *If a position p of t is reduction-complete under ϕ , then $s\sigma \gtrsim_{\text{Expd}_R(s \gtrsim t[\phi], p)}^{\square} \circ \leftarrow_{\mathcal{R}} t\sigma$.*

Proof. We here show only (a) since (b) can be proved analogously to (a). Suppose that a position p of s is reduction-complete under ϕ . Since p is reduction-complete under ϕ , we have that $l\theta \equiv s|_p\sigma$ and $\mathcal{M} \models \psi\theta$ for some rewrite rule $l \rightarrow r[\psi] \in \mathcal{R}$ and substitution θ grounding l . We now assume w.l.o.g. that $\text{Var}(l) \cap \text{Var}(s|_p) = \emptyset$ and $\text{Dom}(\sigma) \cap \text{Dom}(\theta) = \emptyset$. Since $s|_p\sigma \equiv s|_p\theta\sigma$ and $l\theta \equiv l\theta\sigma$, we can see that $\theta\sigma$ is a unifier of $s|_p$ and l . Thus, there exist an mgu δ of $s|_p$ and l , and a ground substitution ν such that $\theta\sigma = \delta\nu$. Since $\mathcal{M} \models \phi\sigma$ and $\mathcal{M} \models \psi\theta$, we also have that $\mathcal{M} \models \phi\theta\sigma$ and $\mathcal{M} \models \psi\theta\sigma$. Moreover, since $\theta\sigma = \delta\nu$, we have that $\mathcal{M} \models (\phi \wedge \psi)\delta\nu$. By the definition of Expd_L , we have that $(s[r]_p)\delta \gtrsim t\delta[(\phi \wedge \psi)\delta] \in \text{Expd}_L(s \gtrsim t[\phi], p)$. Therefore, $s\sigma \equiv s\theta\sigma \equiv s\delta\nu \equiv s\delta\nu[s|_p\delta\nu]_p \equiv s\delta\nu[l\delta\nu]_p \rightarrow_{\mathcal{R}} s\delta\nu[r\delta\nu]_p \equiv (s[r]_p)\delta\nu \gtrsim_{\text{Expd}_L(s \gtrsim t[\phi], p)}^{\square} t\delta\nu \equiv t\theta\sigma \equiv t\sigma$. $\square$

The next lemma states soundness of each inference rule.

Lemma 5.14. *If $(\mathcal{E}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow}) \vdash_{RI} (\mathcal{E}', \mathcal{H}'_{\rightarrow}, \mathcal{H}'_{\leftarrow})$, then $\gtrsim_{\mathcal{E}}^{\text{mono}} \subseteq (\rightarrow_{\mathcal{R}} \cup \gtrsim_{\mathcal{H}'_{\rightarrow}}^{\text{iso}} \cup \gtrsim_{\mathcal{H}'_{\leftarrow}}^{\text{anti}})^* \circ (\gtrsim_{\mathcal{E}'}^{\text{mono}} \cup \gtrsim_{\mathcal{R}}) \circ (\leftarrow_{\mathcal{R}} \cup \gtrsim_{\mathcal{H}'_{\leftarrow}}^{\text{iso}} \cup \gtrsim_{\mathcal{H}'_{\rightarrow}}^{\text{anti}})^*$ on ground terms.*

Proof. We make a case distinction depending on the applied inference rule.

- *The first rule of Rule-Application.* We prove that $\gtrsim_{\mathcal{E} \setminus \mathcal{E}'}^{\square} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \gtrsim_{\mathcal{E}'}^{\square} \circ \leftarrow_{\mathcal{R}}^*$ since by Lemma 5.8 (c) the inclusion implies that $\gtrsim_{\mathcal{E} \setminus \mathcal{E}'}^{\text{iso}} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \gtrsim_{\mathcal{E}'}^{\text{iso}} \circ \leftarrow_{\mathcal{R}}^*$ and $\gtrsim_{\mathcal{E} \setminus \mathcal{E}'}^{\text{anti}} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \gtrsim_{\mathcal{E}'}^{\text{anti}} \circ \leftarrow_{\mathcal{R}}^*$, which furthermore implies $\gtrsim_{\mathcal{E} \setminus \mathcal{E}'}^{\text{mono}} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \gtrsim_{\mathcal{E}'}^{\text{mono}} \circ \leftarrow_{\mathcal{R}}^*$. Let $u \gtrsim_{\mathcal{E} \setminus \mathcal{E}'}^{\square} v$. By definition, there exist an inequality $s \gtrsim t[\phi] \in \mathcal{E} \setminus \mathcal{E}'$ and a substitution θ grounding s and t such that $u \equiv s\theta$, $v \equiv t\theta$, $\mathcal{Ran}(\theta|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$, and $\mathcal{M} \models \phi\theta$. Then, there also exist a rewrite rule $l \rightarrow r[\psi] \in \mathcal{R}$, a context $C[\]$, and a substitution σ such that $s \equiv C[l\sigma]$, $\mathcal{Ran}(\sigma|_{FV(\psi)}) \subseteq \mathcal{T}(\mathcal{G}, \mathcal{V})$, and $\phi \Rightarrow \psi\sigma$ is $\mathcal{M}$ -valid. Let $X = \{x \in \text{Dom}(\theta) \mid x\theta \in \mathcal{T}(\mathcal{G}, \mathcal{V})\}$. Since $\phi \Rightarrow \psi\sigma$ is $\mathcal{M}$ -valid, $\mathcal{M} \models \phi\theta$, and $\phi\theta \equiv \phi(\theta|_X)$, we can see that $\psi\sigma(\theta|_X)$ is $\mathcal{M}$ -valid. By Lemma 5.12, there exists a substitution δ grounding s and t such that $s\theta \rightarrow_{\mathcal{R}}^* s\delta$, $t\theta \rightarrow_{\mathcal{R}}^* t\delta$, $\mathcal{M} \models \psi\sigma\delta$, and $\theta|_X = \delta|_X$. Here, since $\mathcal{Ran}(\theta|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$, we have that $\phi\theta \equiv \phi(\theta|_X) \equiv \phi(\delta|_X) \equiv \phi\delta$. Since $C[r\sigma] \gtrsim t \in \mathcal{E}'$, $\mathcal{M} \models \phi\theta$, and $\phi\theta \equiv \phi\delta$, we have that $C[r\sigma]\delta \gtrsim_{\mathcal{E}'}^{\square} t\delta$. Therefore, we have that $u \equiv s\theta \rightarrow_{\mathcal{R}}^* s\delta \equiv C[l\sigma]\delta \equiv C\delta[l\sigma\delta] \rightarrow_{\mathcal{R}} C\delta[r\sigma\delta] \equiv C[r\sigma]\delta \gtrsim_{\mathcal{E}'}^{\square} t\delta \leftarrow_{\mathcal{R}}^* t\theta \equiv v$.

- *The second rule of Rule-Application.* As in the previous case, we can show that $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{\square} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}'}^{\square} \circ \leftarrow_{\mathcal{R}}^+$. By Lemma 5.8 (d), we have that $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{iso} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}'}^{iso} \circ \leftarrow_{\mathcal{R}}^+$ and $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{anti} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \succsim_{\mathcal{E}'}^{anti} \circ \leftarrow_{\mathcal{R}}^*$, which implies $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{mono} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}'}^{mono} \circ \leftarrow_{\mathcal{R}}^*$.
- *The first rule of L-Hypothesis-Application.* It suffices to show that $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{iso} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{H}'_{\perp}}^{iso} \circ \succsim_{\mathcal{E}'}^{iso} \circ \leftarrow_{\mathcal{R}}^*$ and $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{anti} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{H}'_{\perp}}^{anti} \circ \succsim_{\mathcal{E}'}^{anti} \circ \leftarrow_{\mathcal{R}}^*$.
 - We first show that $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{iso} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{H}'_{\perp}}^{iso} \circ \succsim_{\mathcal{E}'}^{iso} \circ \leftarrow_{\mathcal{R}}^*$. Let $z \succsim_{\mathcal{E} \setminus \mathcal{E}'}^{iso} w$. By definition, there exist an inequality $s \succsim t [\phi] \in \mathcal{E} \setminus \mathcal{E}'$, a context $D[\]$ isotone under $\top$, and a substitution θ grounding s and t such that $z \equiv D[s\theta]$, $w \equiv D[t\theta]$, $\mathcal{R}an(\theta|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$, and $\mathcal{M} \models \phi\theta$. Then, there also exist an inequality $u \succsim v [\psi] \in \mathcal{H}_{\perp}$, a context $C[\]$ isotone under ϕ , and a substitution σ such that $s \equiv C[u\sigma]$, $\mathcal{R}an(\sigma|_{FV(\psi)}) \subseteq \mathcal{T}(\mathcal{G}, \mathcal{V})$, and $\phi \Rightarrow \psi\sigma$ is $\mathcal{M}$ -valid. Let $X = \{x \in \text{Dom}(\theta) \mid x\theta \in \mathcal{T}(\mathcal{G}, \mathcal{V})\}$. Since $\phi \Rightarrow \psi\sigma$ is $\mathcal{M}$ -valid, $\mathcal{M} \models \phi\theta$, and $\phi\theta \equiv \phi(\theta|_X)$, we can see that $\psi\sigma(\theta|_X)$ is $\mathcal{M}$ -valid. By Lemma 5.12, there exists a substitution δ grounding s and t such that $s\theta \rightarrow_{\mathcal{R}}^* s\delta$, $t\theta \rightarrow_{\mathcal{R}}^* t\delta$, $\mathcal{M} \models \psi\sigma\delta$, and $\theta|_X = \delta|_X$. Here, since $\mathcal{R}an(\theta|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$, we have that $\phi\theta \equiv \phi(\theta|_X) \equiv \phi(\delta|_X) \equiv \phi\delta$. Since $\mathcal{M} \models \phi\theta$ and $\phi\theta \equiv \phi\delta$, it follows from Lemma 5.10 that $C\delta[\]$ is isotone under $\top$. By Lemma 5.9, $D[C\delta[\]]$ is isotone under $\top$. Since $u \succsim v [\psi] \in \mathcal{H}_{\perp}$ and $\mathcal{M} \models \psi\sigma\delta$, we have that $D[C\delta[u\sigma\delta]] \succsim_{\mathcal{H}'_{\perp}}^{iso} D[C\delta[v\sigma\delta]]$. Moreover, since $C[v\sigma] \succsim t [\phi] \in \mathcal{E}'$, $\mathcal{M} \models \phi\theta$, and $\phi\theta \equiv \phi\delta$, we have that $D[C[v\sigma]\delta] \succsim_{\mathcal{E}'}^{iso} D[t\delta]$. Therefore, we have that $z \equiv D[s\theta] \rightarrow_{\mathcal{R}}^* D[s\delta] \equiv D[C[u\sigma]\delta] \equiv D[C\delta[u\sigma\delta]] \succsim_{\mathcal{H}'_{\perp}}^{iso} D[C\delta[v\sigma\delta]] \equiv D[C[v\sigma]\delta] \succsim_{\mathcal{E}'}^{iso} D[t\delta] \leftarrow_{\mathcal{R}}^* D[t\theta] \equiv w$, and hence $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{iso} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{H}'_{\perp}}^{iso} \circ \succsim_{\mathcal{E}'}^{iso} \circ \leftarrow_{\mathcal{R}}^*$.
 - We next show that $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{anti} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{H}'_{\perp}}^{anti} \circ \succsim_{\mathcal{E}'}^{anti} \circ \leftarrow_{\mathcal{R}}^*$. Let $z \succsim_{\mathcal{E} \setminus \mathcal{E}'}^{anti} w$. Then, as in the previous case, we have that $z \equiv D[t\theta] \rightarrow_{\mathcal{R}}^* D[t\delta] \succsim_{\mathcal{E}'}^{anti} D[C[v\sigma]\delta] \equiv D[C\delta[v\sigma\delta]] \succsim_{\mathcal{H}'_{\perp}}^{anti} D[C\delta[u\sigma\delta]] \equiv D[C[u\sigma]\delta] \equiv D[s\delta] \leftarrow_{\mathcal{R}}^* D[s\theta] \equiv w$, where $D[\]$ is a context antitone under $\top$.
- *The second rule of L-Hypothesis-Application.* It suffices to show that $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{iso} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{H}'_{\perp}}^{iso} \circ \succsim_{\mathcal{E}'}^{anti} \circ \leftarrow_{\mathcal{R}}^*$ and $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{anti} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{H}'_{\perp}}^{iso} \circ \succsim_{\mathcal{E}'}^{anti} \circ \leftarrow_{\mathcal{R}}^*$. The proof is analogous to the case where the first rule is applied.
- *R-Hypothesis-Application.* Similarly to the case of L-Hypothesis-Application, if the first rule is applied, then the claim holds since $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{iso} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{H}'_{\perp}}^{iso} \circ \succsim_{\mathcal{E}'}^{iso} \circ \leftarrow_{\mathcal{R}}^*$ and $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{anti} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{H}'_{\perp}}^{anti} \circ \succsim_{\mathcal{E}'}^{anti} \circ \leftarrow_{\mathcal{R}}^*$. If the second rule is applied, then the claim follows since $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{iso} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{H}'_{\perp}}^{anti} \circ \succsim_{\mathcal{E}'}^{iso} \circ \leftarrow_{\mathcal{R}}^*$ and $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{anti} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{H}'_{\perp}}^{iso} \circ \succsim_{\mathcal{E}'}^{anti} \circ \leftarrow_{\mathcal{R}}^*$.
- *Deletion.* Suppose that $s \succsim t [\phi] \in \mathcal{E}$ is removed, i.e., $\mathcal{E} \setminus \mathcal{E}' = \{s \succsim t [\phi]\}$. By Proposition 3.6, $s \succsim t [\phi]$ is an inductive theorem of $\mathcal{R}$. By Lemma 5.5, we have that $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{mono} \subseteq \succsim_{\mathcal{R}}$.
- *L-Expansion.* We show that $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{\square} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \succsim_{\mathcal{E}'}^{\square} \circ \leftarrow_{\mathcal{R}}^*$ since by Lemma 5.8 (c) the inclusion implies that $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{iso} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \succsim_{\mathcal{E}'}^{iso} \circ \leftarrow_{\mathcal{R}}^*$ and $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{anti} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}'}^{anti} \circ \leftarrow_{\mathcal{R}}^+$, which furthermore implies $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{mono} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}'}^{mono} \circ \leftarrow_{\mathcal{R}}^*$. Let $u \succsim_{\mathcal{E} \setminus \mathcal{E}'}^{\square} v$. Then, there exist an inequality $s \succsim t [\phi] \in \mathcal{E} \setminus \mathcal{E}'$ and a ground substitution σ such that $u \equiv s\sigma$, $v \equiv t\sigma$, $\mathcal{R}an(\sigma|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$, $\mathcal{M} \models \phi\sigma$, and a position p of s is reduction-complete under ϕ . By Lemma 5.11, there exists a normalized ground substitution θ such that $s\sigma \rightarrow_{\mathcal{R}}^* s\theta$, $t\sigma \rightarrow_{\mathcal{R}}^* t\theta$, $\mathcal{R}an(\theta|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$, and $\mathcal{M} \models \phi\theta$. By Lemma 5.13 (a), we have that $s\theta \rightarrow_{\mathcal{R}} \succsim_{Expd_L(s \succsim t [\phi], p)}^{\square} t\theta$. Accordingly, we obtain $u \equiv s\sigma \rightarrow_{\mathcal{R}}^* s\theta \rightarrow_{\mathcal{R}} \succsim_{\mathcal{E}'}^{\square} t\theta \leftarrow_{\mathcal{R}}^* t\sigma$, and hence $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{\square} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \succsim_{\mathcal{E}'}^{\square} \circ \leftarrow_{\mathcal{R}}^*$.
- *R-Expansion.* As in the case of L-Expansion, we can show that $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{\square} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}'}^{\square} \circ \leftarrow_{\mathcal{R}}^+$. By Lemma 5.8 (d), we have that $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{iso} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}'}^{iso} \circ \leftarrow_{\mathcal{R}}^+$ and $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{anti} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \succsim_{\mathcal{E}'}^{anti} \circ \leftarrow_{\mathcal{R}}^*$, which implies $\succsim_{\mathcal{E} \setminus \mathcal{E}'}^{mono} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}'}^{mono} \circ \leftarrow_{\mathcal{R}}^*$.

□

The following lemma describes a property of induction hypotheses raised by the L/R-Expansion rules.

Lemma 5.15. *Both of the following statements hold on ground terms:*

- (a) If $(\mathcal{E}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow}) \vdash_{LExp} (\mathcal{E}', \mathcal{H}'_{\rightarrow}, \mathcal{H}'_{\leftarrow})$, then $\succsim_{\mathcal{H}'_{\rightarrow} \setminus \mathcal{H}_{\rightarrow}}^{\square} \subseteq \rightarrow_{\mathcal{R}}^+ \circ \succsim_{\mathcal{E}'}^{\square} \circ \leftarrow_{\mathcal{R}}^*$.

(b) If $(\mathcal{E}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow}) \vdash_{RExp} (\mathcal{E}', \mathcal{H}'_{\rightarrow}, \mathcal{H}'_{\leftarrow})$, then $\succsim_{\mathcal{H}'_{\rightarrow} \setminus \mathcal{H}_{\rightarrow}}^{\square} \subseteq \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}'}^{\square} \circ \leftarrow_{\mathcal{R}}^+$.

Proof. We here show only (a) since (b) can be proved analogously to (a). Assume that $(\mathcal{E}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow}) \vdash_{LExp} (\mathcal{E}', \mathcal{H}'_{\rightarrow}, \mathcal{H}'_{\leftarrow})$. Let $s \succsim_{\mathcal{H}'_{\rightarrow} \setminus \mathcal{H}_{\rightarrow}}^{\square} t$. By definition, there exist an inequality $s' \succsim t' [\phi] \in \mathcal{H}'_{\rightarrow} \setminus \mathcal{H}_{\rightarrow}$ and a ground substitution σ such that $s \equiv s' \sigma$, $t \equiv t' \sigma$, $\mathcal{R}an(\sigma|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$, and $\mathcal{M} \models \phi \sigma$. Since the L-Expansion rule is applied, there exists a position p of s' which is reduction-complete under ϕ . By Lemma 5.11, there exists a normalized ground substitution θ such that $s' \sigma \rightarrow_{\mathcal{R}}^* s' \theta$, $t' \sigma \rightarrow_{\mathcal{R}}^* t' \theta$, $\mathcal{R}an(\theta|_{FV(\phi)}) \subseteq \mathcal{T}(\mathcal{G})$, and $\mathcal{M} \models \phi \theta$. It follows from Lemma 5.13 (a) that $s' \theta \rightarrow_{\mathcal{R}} \circ \succsim_{Expd_L(s' \succsim t' [\phi], p)}^{\square} t' \theta$. Therefore, we have that $s \equiv s' \sigma \rightarrow_{\mathcal{R}}^* s' \theta \rightarrow_{\mathcal{R}} \circ \succsim_{\mathcal{E}'}^{\square} t' \theta \leftarrow_{\mathcal{R}}^* t' \sigma \equiv t$ and thus $s \rightarrow_{\mathcal{R}} \circ \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}'}^{\square}) \circ \leftarrow_2^* t$. $\square$

The following lemma states invariance of two kinds of inclusions.

Lemma 5.16. Let $(\mathcal{E}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow}) \vdash_{RI} (\mathcal{E}', \mathcal{H}'_{\rightarrow}, \mathcal{H}'_{\leftarrow})$, $\rightarrow_1 = \rightarrow_{\mathcal{R}} \cup \succsim_{\mathcal{H}_{\rightarrow}}^{iso} \cup \succsim_{\mathcal{H}_{\leftarrow}}^{anti}$, $\rightarrow_{1'} = \rightarrow_{\mathcal{R}} \cup \succsim_{\mathcal{H}'_{\rightarrow}}^{iso} \cup \succsim_{\mathcal{H}'_{\leftarrow}}^{anti}$, $\leftarrow_2 = \leftarrow_{\mathcal{R}} \cup \succsim_{\mathcal{H}_{\rightarrow}}^{iso} \cup \succsim_{\mathcal{H}_{\leftarrow}}^{anti}$, and $\leftarrow_{2'} = \leftarrow_{\mathcal{R}} \cup \succsim_{\mathcal{H}'_{\rightarrow}}^{iso} \cup \succsim_{\mathcal{H}'_{\leftarrow}}^{anti}$. Then, both of the following statements hold on ground terms:

- (a) If $\rightarrow_1 \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_2^*$, then $\rightarrow_{1'} \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_{1'}^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}'}^{mono}) \circ \leftarrow_{2'}^*$.
- (b) If $\leftarrow_2 \subseteq \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_2^* \circ \leftarrow_{\mathcal{R}}$, then $\leftarrow_{2'} \subseteq \rightarrow_{1'}^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_{2'}^* \circ \leftarrow_{\mathcal{R}}$.

Proof. (a) It suffices to show the following two statements on ground terms:

- (i) If $\succsim_{\mathcal{H}_{\rightarrow}}^{iso} \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_2^*$, then $\succsim_{\mathcal{H}'_{\rightarrow}}^{iso} \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_{1'}^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_{2'}^*$.
- (ii) If $\succsim_{\mathcal{H}_{\leftarrow}}^{anti} \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_2^*$, then $\succsim_{\mathcal{H}'_{\leftarrow}}^{anti} \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_{1'}^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_{2'}^*$.

We first show (i). Suppose that $\succsim_{\mathcal{H}_{\rightarrow}}^{iso} \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_2^*$, and let $s \succsim_{\mathcal{H}'_{\rightarrow}}^{iso} t$. Since $\mathcal{H}_{\rightarrow} \subseteq \mathcal{H}'_{\rightarrow}$ and $\mathcal{H}_{\leftarrow} \subseteq \mathcal{H}'_{\leftarrow}$ by the definition of the inference rules, we have that either $s \succsim_{\mathcal{H}_{\rightarrow}}^{iso} t$ or $s \succsim_{\mathcal{H}'_{\rightarrow} \setminus \mathcal{H}_{\rightarrow}}^{iso} t$.

- Consider the case where $s \succsim_{\mathcal{H}_{\rightarrow}}^{iso} t$. By assumption, $s \rightarrow_{\mathcal{R}} \circ \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_2^* t$. Since $\mathcal{H}_{\rightarrow} \subseteq \mathcal{H}'_{\rightarrow}$ and $\mathcal{H}_{\leftarrow} \subseteq \mathcal{H}'_{\leftarrow}$, we have that $s \rightarrow_{\mathcal{R}} \circ \rightarrow_{1'}^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_{2'}^* t$. Now, by Lemma 5.14, we have that $\succsim_{\mathcal{E}}^{mono} \subseteq \rightarrow_1^* \circ (\leftarrow_{\mathcal{E}'}^* \cup \succsim_{\mathcal{R}}) \circ \leftarrow_2^*$. Hence, we have that $s \rightarrow_{\mathcal{R}} \circ \rightarrow_{1'}^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}'}^{mono}) \circ \leftarrow_{2'}^* t$.
- Consider the remaining case where $s \succsim_{\mathcal{H}'_{\rightarrow} \setminus \mathcal{H}_{\rightarrow}}^{iso} t$. Since the L-Expansion rule is the only rule which makes $\mathcal{H}'_{\rightarrow} \setminus \mathcal{H}_{\rightarrow}$ non-empty, the L-Expansion rule is applied in the derivation. By Lemmas 5.15 (a) and 5.8 (c), we have that $s \rightarrow_{\mathcal{R}}^+ \circ \succsim_{\mathcal{E}'}^{iso} \circ \leftarrow_{\mathcal{R}}^* t$.

Next, we show (ii). Suppose that $\succsim_{\mathcal{H}_{\leftarrow}}^{anti} \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_2^*$, and let $s \succsim_{\mathcal{H}'_{\leftarrow}}^{anti} t$. Since $\mathcal{H}_{\leftarrow} \subseteq \mathcal{H}'_{\leftarrow}$ and $\mathcal{H}_{\rightarrow} \subseteq \mathcal{H}'_{\rightarrow}$ by the definition of the inference rules, we have two cases to consider: $s \succsim_{\mathcal{H}_{\leftarrow}}^{anti} t$ or $s \succsim_{\mathcal{H}'_{\leftarrow} \setminus \mathcal{H}_{\leftarrow}}^{anti} t$. The first case is analogous to (i). As for the second case, since we can see that the R-Expansion rule is applied, we have that $s \rightarrow_{\mathcal{R}}^+ \circ \succsim_{\mathcal{E}'}^{anti} \circ \leftarrow_{\mathcal{R}}^* t$ by Lemmas 5.15 (b) and 5.8 (d).

(b) It suffices to show the following two statements on ground terms:

- (iii) If $\succsim_{\mathcal{H}_{\leftarrow}}^{iso} \subseteq \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_2^* \circ \leftarrow_{\mathcal{R}}$, then $\succsim_{\mathcal{H}'_{\leftarrow}}^{iso} \subseteq \rightarrow_{1'}^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_{2'}^* \circ \leftarrow_{\mathcal{R}}$.
- (iv) If $\succsim_{\mathcal{H}_{\rightarrow}}^{anti} \subseteq \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_2^* \circ \leftarrow_{\mathcal{R}}$, then $\succsim_{\mathcal{H}'_{\rightarrow}}^{anti} \subseteq \rightarrow_{1'}^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_{2'}^* \circ \leftarrow_{\mathcal{R}}$.

We first show (iii). Suppose that $\succsim_{\mathcal{H}_{\leftarrow}}^{iso} \subseteq \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_2^* \circ \leftarrow_{\mathcal{R}}$, and let $s \succsim_{\mathcal{H}'_{\leftarrow}}^{iso} t$. Since $\mathcal{H}_{\leftarrow} \subseteq \mathcal{H}'_{\leftarrow}$ and $\mathcal{H}_{\rightarrow} \subseteq \mathcal{H}'_{\rightarrow}$, we have that two cases to consider: $s \succsim_{\mathcal{H}_{\leftarrow}}^{iso} t$ or $s \succsim_{\mathcal{H}'_{\leftarrow} \setminus \mathcal{H}_{\leftarrow}}^{iso} t$. For the first case, we have that $s \rightarrow_{1'}^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_{2'}^* \circ \leftarrow_{\mathcal{R}} t$. For the second case, since the L-Expansion rule is applied, it follows from Lemmas 5.15 (b) and 5.8 (d) that $s \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}'}^{iso} \circ \leftarrow_{\mathcal{R}}^+ t$.

Next, we show (iv). Suppose that $\succsim_{\mathcal{H}_{\rightarrow}}^{anti} \subseteq \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \succsim_{\mathcal{R}} \cup \succsim_{\mathcal{E}}^{mono}) \circ \leftarrow_2^* \circ \leftarrow_{\mathcal{R}}$, and let $s \succsim_{\mathcal{H}'_{\rightarrow}}^{anti} t$. We have two cases to consider: $s \succsim_{\mathcal{H}_{\rightarrow}}^{anti} t$ or $s \succsim_{\mathcal{H}'_{\rightarrow} \setminus \mathcal{H}_{\rightarrow}}^{anti} t$. The first case is analogous to (iii). As for the second case, since the L-Expansion rule is applied, it follows from Lemmas 5.15 (a) and 5.8 (c) that $s \rightarrow_{\mathcal{R}}^* \circ \succsim_{\mathcal{E}'}^{anti} \circ \leftarrow_{\mathcal{R}}^+ t$. $\square$

The next lemma states invariance of well-foundedness.

Lemma 5.17. *Let $(\mathcal{E}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow}) \vdash_{RI} (\mathcal{E}', \mathcal{H}'_{\rightarrow}, \mathcal{H}'_{\leftarrow})$. If $(\rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}'_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}'_{\leftarrow}}) \cup (\leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}'_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}'_{\rightarrow}})^{-1}$ is well-founded on ground terms, then so is $(\rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\leftarrow}}) \cup (\leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\rightarrow}})^{-1}$.*

Proof. If an inference rule other than L-Expansion or R-Expansion is applied, the claim trivially holds since $\mathcal{H}'_{\rightarrow} = \mathcal{H}_{\rightarrow}$ and $\mathcal{H}'_{\leftarrow} = \mathcal{H}_{\leftarrow}$. Let us consider the other case where the L-Expansion or R-Expansion rule is applied. Let $\mathcal{R}_1 = \{u \rightarrow v [\psi] \mid u \gtrsim v [\psi] \in \mathcal{H}'_{\rightarrow}\}$ and $\mathcal{R}_2 = \{v \rightarrow u [\psi] \mid u \gtrsim v [\psi] \in \mathcal{H}'_{\leftarrow}\}$. By the definition of the inference rules, we can see that $\mathcal{R} \cup \mathcal{R}_1 \cup \mathcal{R}_2$ is terminating, and hence $\rightarrow_{\mathcal{R} \cup \mathcal{R}_1 \cup \mathcal{R}_2}$ is well-founded on ground terms. Now, it follows from the definition that $(\rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}'_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}'_{\leftarrow}}) \cup (\leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}'_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}'_{\rightarrow}})^{-1} \subseteq (\rightarrow_{\mathcal{R}} \cup \rightarrow_{\mathcal{R}_1} \cup \rightarrow_{\mathcal{R}_2}) \cup (\leftarrow_{\mathcal{R}} \cup \leftarrow_{\mathcal{R}_2} \cup \leftarrow_{\mathcal{R}_1})^{-1} = \rightarrow_{\mathcal{R}} \cup \rightarrow_{\mathcal{R}_1} \cup \rightarrow_{\mathcal{R}_2} = \rightarrow_{\mathcal{R} \cup \mathcal{R}_1 \cup \mathcal{R}_2}$. Therefore, $(\rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\leftarrow}}) \cup (\leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\rightarrow}})^{-1}$ is well-founded on ground terms. $\square$

Lemmas 5.14, 5.16, and 5.17 can be generalized for arbitrary lengths of derivations as follows.

Lemma 5.18. *Let $(\mathcal{E}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow}) \vdash_{RI}^* (\mathcal{E}', \mathcal{H}'_{\rightarrow}, \mathcal{H}'_{\leftarrow})$, $\rightarrow_1 = \rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\leftarrow}}$, $\rightarrow_{1'} = \rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}'_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}'_{\leftarrow}}$, $\leftarrow_2 = \leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\rightarrow}}$, and $\leftarrow_{2'} = \leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}'_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}'_{\rightarrow}}$. Then, all of the following statements hold on ground terms:*

- (a) $\gtrsim_{\mathcal{E}}^{\text{mono}} \subseteq \rightarrow_1^* \circ (\gtrsim_{\mathcal{E}'}^{\text{mono}} \cup \gtrsim_{\mathcal{R}}) \circ \leftarrow_2^*$.
- (b) If $\rightarrow_1 \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \gtrsim_{\mathcal{R}} \cup \gtrsim_{\mathcal{E}}^{\text{mono}}) \circ \leftarrow_2^*$, then $\rightarrow_{1'} \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_{1'}^* \circ (\leftarrow_{\mathcal{R}}^* \cup \gtrsim_{\mathcal{R}} \cup \gtrsim_{\mathcal{E}'}^{\text{mono}}) \circ \leftarrow_{2'}^*$.
- (c) If $\leftarrow_2 \subseteq \rightarrow_1^* \circ (\leftarrow_{\mathcal{R}}^* \cup \gtrsim_{\mathcal{R}} \cup \gtrsim_{\mathcal{E}}^{\text{mono}}) \circ \leftarrow_2^* \circ \leftarrow_{\mathcal{R}}$, then $\leftarrow_{2'} \subseteq \rightarrow_{1'}^* \circ (\leftarrow_{\mathcal{R}}^* \cup \gtrsim_{\mathcal{R}} \cup \gtrsim_{\mathcal{E}'}^{\text{mono}}) \circ \leftarrow_{2'}^* \circ \leftarrow_{\mathcal{R}}$.
- (d) If $\rightarrow_1 \cup \rightarrow_2$ is well-founded, then so is $\rightarrow_{1'} \cup \rightarrow_{2'}$.

Proof. We prove all the claims by induction on the length of the derivation $(\mathcal{E}, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow}) \vdash_{RI}^* (\mathcal{E}', \mathcal{H}'_{\rightarrow}, \mathcal{H}'_{\leftarrow})$: (a) follows from Lemma 5.14, (b) from Lemma 5.16 (a), (c) from Lemma 5.16 (b), and (d) from Lemma 5.17. $\square$

Finally, the following theorem states soundness of our framework. In the proof of this theorem, the binary relations $\rightarrow_1$, $\rightarrow_2$, $\rightarrow_3$, and $\rightarrow_4$ in Theorem 5.2 are assigned to concrete binary relations as follows: $\rightarrow_1 = \rightarrow_{\mathcal{R}}$, $\rightarrow_2 = \rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\leftarrow}}$, $\rightarrow_3 = (\leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\rightarrow}})^{-1}$, and $\rightarrow_4 = \gtrsim_{\mathcal{R}}$.

Theorem 5.19. *Suppose that $\mathcal{R}$ is ground convergent, sufficiently complete w.r.t. interpretable terms, and locally sound w.r.t. $\mathcal{M}$. Let $\gtrsim : \tau \times \tau$ be a binary predicate symbol in $\mathcal{P}$ transitive under $\mathcal{M}$ and $\mathcal{E}$ be a set of constrained inequalities. If $(\mathcal{E}, \emptyset, \emptyset) \vdash_{RI}^* (\emptyset, \mathcal{H}_{\rightarrow}, \mathcal{H}_{\leftarrow})$, then every inequality in $\mathcal{E}$ is an inductive theorem of $\mathcal{R}$.*

Proof. By Lemma 5.5, it suffices to show that $\gtrsim_{\mathcal{E}}^{\text{mono}} \subseteq \gtrsim_{\mathcal{R}}$ on ground terms. By Lemma 5.18 (a), we have that $\gtrsim_{\mathcal{E}}^{\text{mono}} \subseteq (\rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\leftarrow}})^* \circ \gtrsim_{\mathcal{R}} \circ (\leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\rightarrow}})^*$. Now we observe the following facts:

- We clearly have that $\rightarrow_{\mathcal{R}} \subseteq \rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\leftarrow}}$ and $\leftarrow_{\mathcal{R}} \subseteq \leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\rightarrow}}$.
- By Lemma 5.18 (d), $(\rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\leftarrow}}) \cup (\leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\rightarrow}})^{-1}$ is well-founded.
- It follows from Lemma 5.18 (b) that $\rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\leftarrow}} \subseteq \rightarrow_{\mathcal{R}} \circ (\rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\leftarrow}})^* \circ (\leftarrow_{\mathcal{R}}^* \cup \gtrsim_{\mathcal{R}}) \circ (\leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\rightarrow}})^*$.
- It follows from Lemma 5.18 (c) that $\leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\rightarrow}} \subseteq (\rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\leftarrow}})^* \circ (\leftarrow_{\mathcal{R}}^* \cup \gtrsim_{\mathcal{R}}) \circ (\leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\rightarrow}})^* \circ \leftarrow_{\mathcal{R}}$.

It follows from Theorem 5.2 that $(\rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\leftarrow}})^* \subseteq (\leftarrow_{\mathcal{R}} \cup \gtrsim_{\mathcal{R}})^*$ and $(\leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\rightarrow}})^* \subseteq (\leftarrow_{\mathcal{R}} \cup \gtrsim_{\mathcal{R}})^*$, and hence

$$\begin{aligned} \gtrsim_{\mathcal{E}}^{\text{mono}} &\subseteq (\rightarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\rightarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\leftarrow}})^* \circ \gtrsim_{\mathcal{R}} \circ (\leftarrow_{\mathcal{R}} \cup \overset{iso}{\sim}_{\mathcal{H}_{\leftarrow}} \cup \overset{anti}{\sim}_{\mathcal{H}_{\rightarrow}})^* \\ &\subseteq (\leftarrow_{\mathcal{R}} \cup \gtrsim_{\mathcal{R}})^* \circ \gtrsim_{\mathcal{R}} \circ (\leftarrow_{\mathcal{R}} \cup \gtrsim_{\mathcal{R}})^* \\ &\subseteq (\leftarrow_{\mathcal{R}}^* \circ \gtrsim_{\mathcal{R}} \circ \leftarrow_{\mathcal{R}}^*)^+. \end{aligned}$$

Since $(\leftarrow_{\mathcal{R}}^* \circ \gtrsim_{\mathcal{R}} \circ \leftarrow_{\mathcal{R}}^*)^+ = \gtrsim_{\mathcal{R}}$ by Lemma 5.6, we have that $\gtrsim_{\mathcal{E}}^{\text{mono}} \subseteq \gtrsim_{\mathcal{R}}$. $\square$

6. Automation

To automate applying inference rules of an RI framework, one needs to prepare a strategy designed for the framework. For this purpose, in Section 6.1, after briefly recalling some known strategies, we discuss some issues to be resolved to modify existing strategies for our framework directly handling inequalities. Based on some observations, we propose a strategy for our framework. In Section 6.2, we evaluate performance of the strategy through experiments.

6.1. Strategies for applying inference rules

In general, there exists no versatile strategy appropriate for proving any conjectures. For this reason, strategies that can try multiple proof processes are employed in many RI methods. For instance, the method in [30] adopts backtracking mechanism. In [38], a multi-context approach for RI is proposed to simulate multiple proof processes in one process. In this paper, to successfully prove as many conjectures as possible that are inductive theorems, we consider a strategy with a nondeterministic nature that covers several derivations simultaneously.

On the other hand, the order of application of inference rules is basically common to all RI methods, which is described as follows:

- (1) Apply the Simplification rule (corresponding to our Rule-Application and L/R-Hypothesis-Application rules) as much as possible;
- (2) Apply the Deletion and related rules as much as possible;
- (3) Apply the Expansion rule once before going back to (1).

One may start from (3), skipping the first execution of (1) and (2). Existing strategies in [37, 38, 34] actually follow this order. While the strategy presented in [30] is defined differently from the above and selects inference rules on the basis of their priorities, the resulting behavior basically corresponds to the order above.

In [34], the strategy proposed in [37] is slightly modified and more precisely described. For example, application of the Simplification rule is designed to use induction hypotheses in preference to rewrite rules. We adapt this one to our setting. A naive adaptation (with a slight modification), which we here call the *naive strategy*, can be described as the repeated execution of the following atomic operations until no conjecture remains:

Simpl: Apply the Rule-Application and L/R-Hypothesis-Application rules as much as possible, where the L/R-Hypothesis-Application rules are applied in preference to the Rule-Application rule. If a conjecture has multiple applicable positions, we select the leftmost outermost position. We do not use rewrite rules in $\mathcal{R}|_{\mathcal{G}}$.

Del: Apply the Deletion rule as much as possible.

Expd: Apply the L/R-Expansion rules once for each applicable conjecture. We apply the rule at the leftmost outermost reduction-complete position such that the subterm at the position has the form $f(t_1, \dots, t_n)$ with $f \in \text{Root}(\mathcal{R}|_{\mathcal{F}})$ and $t_1, \dots, t_n \in \mathcal{T}((\mathcal{F} \cup \mathcal{G}) \setminus \text{Root}(\mathcal{R}|_{\mathcal{F}}), \mathcal{V})$, where $\text{Root}(\mathcal{R}|_{\mathcal{F}}) = \{\text{root}(l) \mid l \rightarrow r [\phi] \in \mathcal{R}|_{\mathcal{F}}\}$. It is arbitrary which of the left- and right-hand sides is expanded.

Unfortunately, this naive adaptation seems not to be suitable for our framework directly handling inequalities without change. This is because we should not thoughtlessly apply induction hypotheses. Since induction hypotheses are inequalities in our framework, application of induction hypotheses may cause approximation of conjectures, i.e., the application is not always complete, as seen in Example 4.10. To successfully prove the constrained inequality (15) in Example 4.10 to be an inductive theorem of $\mathcal{R}_{\text{pow}}$, we must apply the L-Expansion rule at least twice before applying the L/R-Hypothesis-Application rules.

Example 6.1. Recall $\mathcal{R}_{\text{pow}}$ and the constrained inequality (15) in Example 4.10. As already seen in Example 4.10, if we apply an induction hypothesis after using the L-Expand rule only once, we obtain an incomplete derivation with an inequality (16b') that is not inductively valid. On the other hand, if we apply an induction hypothesis after using the L-Expand rule twice, a successful derivation is obtained as follows:

$$\begin{aligned}
 & \{ (16) \text{ pow}(s^2(0), n) \gtrsim n \mid n \gtrsim 0 \}, \emptyset, \emptyset \\
 & \vdash_{\text{LExp}} \\
 & \left(\left\{ \begin{array}{l} (16a) \quad s(0) \gtrsim n \mid n \gtrsim 0 \wedge n \not\gtrsim s(0), \\ (16b) \quad s^2(0) \times \text{pow}(s^2(0), p(n)) \gtrsim n \mid n \gtrsim 0 \wedge n \gtrsim s(0) \end{array} \right\}, \{ (16) \}, \emptyset \right)
 \end{aligned}$$

$$\begin{aligned}
& \vdash_{LExp} \left(\left(\begin{array}{l} (16a) \quad s(0) \gtrsim n \ [n \gtrsim 0 \wedge n \not\gtrsim s(0)], \\ (16b-1) \quad s^2(0) \times s(0) \gtrsim n \ [n \gtrsim 0 \wedge n \gtrsim s(0) \wedge p(n) \not\gtrsim s(0)], \\ (16b-2) \quad s^2(0) \times (s^2(0) \times \text{pow}(s^2(0), p^2(n))) \gtrsim n \ [n \gtrsim 0 \wedge n \gtrsim s(0) \wedge p(n) \gtrsim s(0)] \end{array} \right), \{(16), (16b)\}, \emptyset \right) \\
& \vdash_{Del}^* ((16b-2) \ s^2(0) \times (s^2(0) \times \text{pow}(s^2(0), p^2(n))) \gtrsim n \ [n \gtrsim 0 \wedge n \gtrsim s(0) \wedge p(n) \gtrsim s(0)], \{(16), (16b)\}, \emptyset) \\
& \vdash_{LHypoApp} ((16b-2') \ s^2(0) \times p(n) \gtrsim n \ [n \gtrsim 0 \wedge n \gtrsim s(0) \wedge p(n) \gtrsim s(0)], \{(16), (16b)\}, \emptyset) \\
& \vdash_{Del} (\emptyset, \{(16), (16b)\}, \emptyset).
\end{aligned}$$

Note that the newer induction hypothesis (16b) is applied to obtain (16b-2') from (16b-2). If we apply (16) instead of (16b), an invalid conjecture $s^2(0) \times (s^2(0) \times p^2(n)) \gtrsim n \ [n \gtrsim 0 \wedge n \gtrsim s(0) \wedge p(n) \gtrsim s(0)]$ will be derived.

From Example 6.1, we can see that strategies for our framework should be able to postpone application of the L/R-Hypothesis-Application rules until the L/R-Expansion rules are applied multiple times. Moreover, we should prioritize newly obtained induction hypotheses in the L/R-Hypothesis-Application rules.

Now we design a strategy for our framework based on these observations. We take a simple approach of making the execution of the operation **Simpl** optional, which enables us to postpone application of induction hypotheses. Prioritization of newer induction hypotheses is already realized by the condition in **Simpl** that the leftmost outermost position is selected. Now we define the *modified strategy* by using the above atomic operations as follows: repeatedly apply the operations **Simpl**, **Del**, and **Expd** in this order, where **Simpl** can be skipped. While this strategy takes a higher cost than the native one to execute due to its flexibility, it may avoid failure of proofs caused by too early application of induction hypotheses. Performance of the strategy will be evaluated in the next subsection.

6.2. Experiments

We implemented the proposed framework and both the “naive” and “modified” strategies to evaluate performance of the strategy.⁸ Experiments are conducted on a machine running Linux equipped with an Intel Core i5 CPU at 3.20 GHz with 8 GB RAM. We provide 16 benchmarks⁸ consisting of some conjectures, which are prepared for the experiments. We set the timeout for each test to 300 seconds. To solve validity and satisfiability of constraints, we use the SMT solver Z3 [13]. Termination of constrained TRSs is verified based on [36], and the other properties are verified in the way mentioned in Section 2.4, while we judged ground confluence of $\mathcal{R}_{arith}$ by sight.

Since the strategies are described in a nondeterministic way, a proof process in our implementation proceeds with constructing a tree as in Figure 2. Each operation specified in the strategy is simultaneously executed to all the leaves, and all possibilities are tried in parallel. Nondeterminicity in our strategies is resolved as follows:

- In the modified strategy, since the execution of **Simpl** is optional, each derivation is forked into two at the beginning of each repetition of the strategy cycle.
- In the operation **Expd** of both strategies, if there exist multiple combinations of applicable conjectures and directions (deciding which of the L/R-Expansion rules is applied), then we try all the combinations retaining termination of $\mathcal{R} \cup \overrightarrow{\mathcal{H}_\rightarrow} \cup \overleftarrow{\mathcal{H}_\leftarrow}$.

To avoid meaningless inferences, the following leaves are ignored: (a) leaves containing an obviously invalid conjecture (i.e., a conjecture $s \gtrsim t [\phi]$ such that $s, t \in \mathcal{T}(\mathcal{G}, \mathcal{V})$ and $\phi \Rightarrow s \gtrsim t$ is not $\mathcal{M}$ -valid) and (b) leaves that do not change by any operation. If all the leaves are ignored, then the proof process is aborted. If we obtain a leaf of the form $(\emptyset, \mathcal{H}_\rightarrow, \mathcal{H}_\leftarrow)$, then the proof process halts successfully.

Table 1 shows the results of the experiments. The column labeled $|\mathcal{E}_0|$ refers to the number of initial conjectures. The labels “Naive” and “Modif.” denote the “naive strategy” and the “modified strategy” defined in Section 6.1, respectively. In the “Results” column, “failed” indicates that the proof process was aborted because all the leaves

⁸ Available at <http://www.trs.cm.is.nagoya-u.ac.jp/crisys/inequality/>.

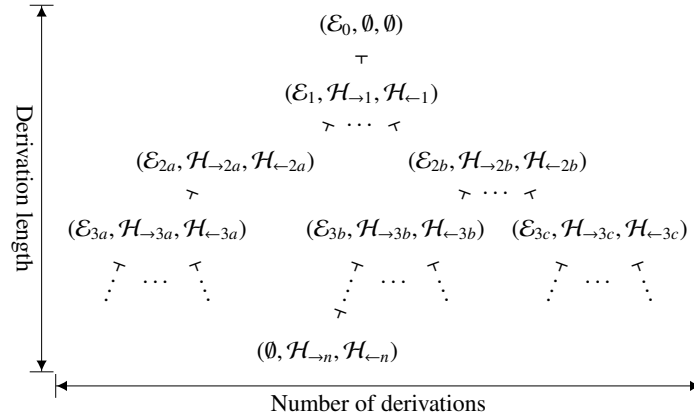
Figure 2. A tree of proof processes with a successful derivation $(\mathcal{E}_0, \emptyset, \emptyset) \vdash \dots \vdash (\emptyset, H_{\rightarrow n}, H_{\leftarrow n})$.

Table 1. Experimental results.

Benchmark	$ \mathcal{E}_0 $	Result		Runtime (in sec.)		#Derivations		Length		#Repetitions	
		Naive	Modif.	Naive	Modif.	Naive	Modif.	Naive	Modif.	Naive	Modif.
1 (Ex. 4.9)	1	proved	proved	0.78	0.99	1	4	4	4	2	2
2 (Ex. 6.1)	1	failed	proved	(0.94)	2.06	(1)	6	-	6	(2)	3
3	1	timeout	timeout	timeout	timeout	(32)	(138)	-	-	(6)	(4)
4 (Ex. 4.7)	6	failed	proved	(2.94)	3.29	(1)	4	-	27	(2)	2
5	2	proved	proved	1.20	1.54	1	4	9	7	2	2
6	2	proved	proved	1.65	1.73	1	4	14	10	2	2
7	2	proved	proved	2.03	1.91	1	4	14	10	2	2
8 (Ex. 4.8)	1	proved	proved	0.52	0.65	1	4	9	9	2	2
9	3	failed	stack overflow	(0.78)	(18.69)	(1)	(32)	-	-	(2)	(16)
10	1	timeout	proved	timeout	29.00	(64)	44	-	10	(7)	3
11	1	timeout	timeout	timeout	timeout	(128)	(128)	-	-	(8)	(4)
12	1	failed	proved	(1.07)	2.08	(1)	6	-	6	(3)	3
13	2	timeout	timeout	timeout	timeout	(102)	(226)	-	-	(4)	(3)
14	1	proved	proved	0.75	1.01	1	4	4	4	2	2
15	1	proved	proved	0.92	1.35	1	4	6	6	2	2
16	1	proved	proved	2.54	3.08	1	4	9	9	2	2

are ignored, and “stack overflow” indicates the proof process was aborted due to stack overflow error caused by too much growth of the size of conjectures. The column “#Derivations” shows the number of derivations (i.e., the number of leaves in Figure 2) generated by the time a successful derivation was obtained. The column “Length” shows the length of the obtained successful derivation (i.e., the height of tree in Figure 2). The column “#Repetitions” shows the number of times the strategy cycle was repeated. The numbers in the parentheses are the values at the time the proof process was aborted due to timeout, failure, or stack overflow error.

From the results of the experiments, we can see that the modified strategy is more appropriate than the naive one as a strategy for proving inequalities. Actually, the modified strategy proved some benchmarks that could not be proved in the naive one. Moreover, all the benchmarks proved in the naive strategy were also successfully proved in the modified one without taking a long additional time. On the other hand, improvement of the operation **Simpl** turned out to be necessary, because it was a major cause of timeout in some benchmarks that the strategy failed to select an appropriate lemma from multiple candidates. To make sure that inductively valid conjectures are successfully proved, we need a strategy with more flexibility on selecting lemmas.

7. Related work

In this section, we describe related work for proving inductive theorems in terms of inequality handling. We focus on the induction principles on which related techniques are based: *explicit induction* and *implicit induction*.

Explicit induction frameworks (e.g., [10, 11]) perform reasoning on the basis of an induction scheme which is explicitly given. They are basically powerful as long as an induction scheme is provided, and some of those frameworks can be used to prove inductive validity of not only equalities but also inequalities, as seen in [12, 11]. However, the fact that those frameworks essentially need a concrete induction scheme can be fatal in the case where the induction scheme is not provided or not obvious. Unlike our framework, it is not sufficient to confirm termination of the rewriting system.

Implicit induction frameworks prove inductive theorems by a *proof by consistency* principle. Inductionless induction [32, 22] and RI [35] are known to be implicit induction principles. Our framework is based on the RI principle and therefore performs implicit induction. Related RI-based techniques working on constrained systems have been developed in [7, 37, 16, 30]. Inductionless-induction-based frameworks working on constrained systems can be found in [19]. Unlike explicit induction frameworks, some implicit induction frameworks such as [37] including our framework require termination but not a concrete induction scheme for reasoning. Moreover, it is known that implicit induction frameworks, where conjectures are processed in a non-hierarchical way, have some advantages of handling mutually recursive functions [6].

To handle inequalities in the previous implicit-induction-based frameworks, one has to reduce inequalities to equations since the existing techniques do not support direct handling of inequalities. As described in Section 1, it is sometimes difficult to prove that equations reduced from inequalities are inductive theorems because of difficulties with making use of transitivity. Note that this situation does not improve even if one uses a framework in, e.g., [9] that supports not only atomic equations but also *clauses*. Although one can use clauses describing transitivity of a binary predicate and monotonicity of a context as lemmas, substantial difficulties are still not resolved.

Many RI-based methods have been implemented as automated theorem provers. SPIKE [8]⁹ and NICE [39]¹⁰ are implementations of implicit induction in conditional theories, but they do not support constrained rewriting. Sail2 [16] and Ctrl [31]¹¹ performs rewriting induction based on constrained rewriting in the same way as ours. These tools are mainly designed for equational reasoning but not for verification of inequalities, and thus they have substantial difficulties in proving inequalities as mentioned above. Actually, for example, Ctrl does not succeed in proving inductive validity of the equation (2) reduced from an inequality (1) in Section 1, returning “Maybe”.

8. Conclusion

In this paper, we have introduced the notion of constrained inequalities and proposed an RI framework for directly proving inductive validity of constrained inequalities. The framework works on constrained TRSs with ground convergence, sufficient completeness w.r.t. interpretable terms, and local soundness w.r.t. the associated structure. Our approach directly handles constrained inequalities and enables us to verify some inequalities that are hard to deal with in the existing RI frameworks. A strategy for applying inference rules was designed for our framework, and it was shown to be reasonable for our framework via some experiments.

One of the directions for future work is to improve strategies and the implementation. As mentioned in Section 6.2, we must improve the way to apply inductive hypotheses so that appropriate lemmas are selected. Moreover, improvement of execution efficiency is an important issue to be tackled. The current implementation constructs a process tree as in Figure 2, and it can easily cause explosion of the number of derivation branches. To avoid this explosion, one may construct a strategy adopting backtracking as in [30]. Adaptation of multi-context approaches [38] to our framework is also considered a promising direction.

Another direction for future work is automation of lemma generation. As seen in Example 4.7, we usually have to add appropriate lemmas. While several generation techniques for constrained systems have been presented in [34, 30], they seem ineffective out of the context of equality verification of programs. Therefore, other generation frameworks suitable for inequalities must be investigated.

⁹ A successor version developed by Sorin Stratulat is available at <https://sites.google.com/site/sorinica/research/spike/>.

¹⁰ Available at <http://www-sop.inria.fr/coprin/urso/logiciels/>.

¹¹ Available at <http://cl-informatik.uibk.ac.at/software/ctrl/>.

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Differences from Our PPDP'16 Paper

Differences from our PPDP'16 paper are summarized as follows:

- We improve the Deletion rule to strictly enhance the power of the framework.
- We develop a strategy for applying inference rules and evaluate their performance through experiments.
- All the proofs for soundness of our inference rules are included.
- We formalize the framework in the context of many-sorted systems because it is difficult to achieve sufficient completeness in single-sorted systems which handle not only the integers but also other data structures.
- To automate proving ground confluence, we slightly modify the examples by removing interpretable function symbols from the left-hand sides of rewrite rules for uninterpreted function symbols. This change does not affect discussions in the paper.
- We show how to prove assumed properties (termination, ground confluence, sufficient completeness, local soundness) by using existing techniques.

The first five above are listed in the introduction of this version.